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Optimization of the drone-assisted pickup and delivery problem

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ABSTRACT

The proposition of utilizing unmanned aerial vehicles (UAVs), colloquially known as drones, for the purpose of delivery was ushered into the mainstream by Amazon in 2013. Since then, UAVs have quickly gained traction as a feasible option for last-mile operations in transport and logistics. In the context of this research, we construct a mathematical framework aiming to depict the collaborative interaction between trucks and drones in coordinating pickup and delivery tasks with the objective of minimizing operational costs. The drone-assisted pickup and delivery problem (DAPDP) is a variant of the problem in which a single truck departs a depot with parcels and a UAV on board. As the truck picks up packages and makes deliveries, the UAV can also be used to make deliveries to customers near the truck's position. As the unmanned aerial vehicle (UAV) embarks on its delivery task, the truck continues on its route, making further deliveries along the way and retrieving the UAV at another customer location different from the launch point. The model is presented as a mixed integer linear program (MILP), and a novel heuristic solution approach based on the classic Clarke–Wright savings heuristic is proposed. Our heuristic's efficacy is evaluated in comparison to a scenario involving only trucks, with a comprehensive series of numerical experiments conducted to underscore the advantages of incorporating UAVs into the pickup and delivery problem. Finally, we perform a comprehensive sensitivity analysis of key drone parameters in order to demonstrate their impact.

1. Introduction

Unmanned aerial vehicles (UAVs) have been the preserve of military and surveillance applications and have only gained considerable adoption in customer-focused markets in the last five years. This is primarily due to the fact that UAVs have evolved to have sticker prices several orders of magnitude lower than those of delivery trucks. This, coupled with their added advantage of higher average speeds in comparison to delivery trucks, suggests potential savings with respect to operational costs and delivery times.

To satisfy customers' desire for quicker delivery times and lower costs, Amazon began testing UAV parcel delivery in December 2013 (Ponza, 2016). Since then, drone deliveries have become a reality for many companies innovating in that space around the world. In addition to Amazon's *Prime Air* UAV (Fig. 1(a)), other UAV projects include DHL's *Parcelcopter* (Fig. 1(b)) and Google's *Project Wing*. In October 2019, the Federal Aviation Authority (FAA) cleared the way for UPS to operate a drone delivery airline. With the FAA's certification, a UPS drone can operate out of an operator's line of sight, it can fly over people, and it can fly at night. The possibilities abound; from transportation of medical supplies to online deliveries, there are many examples of customer-centric services (UPS, 2019).

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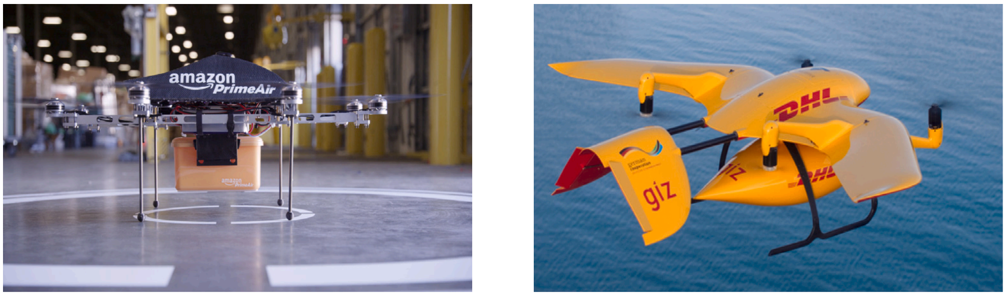
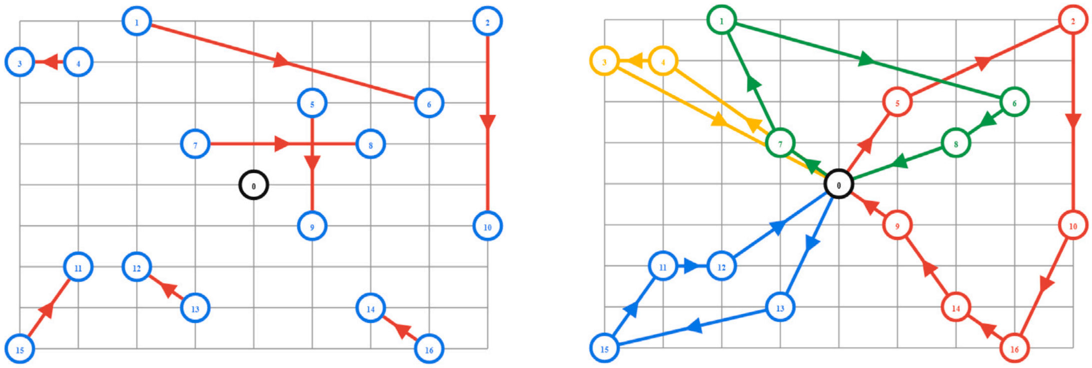


(a) (b)

Fig. 1. Examples of UAVs. (a): *Prime Air* UAV (from: amazon.com). (b) *Parcelcopter* UAV (from: dhl.com).



(a) (b)

Fig. 2. The pickup and delivery problem. (a): Directed edge from pickup to delivery. (b): Vehicle routes.
Source: OR.Tools (2019).

In recent years, UAV-assisted delivery systems have attracted significant attention from companies and governments worldwide, as they offer the potential to enhance last-mile delivery efficiency and reduce operational costs. For instance, in Rwanda, the company Zipline has been using drones to deliver medical supplies to remote areas since 2016 (Tilley, 2016). In China, e-commerce giant JD.com has been testing drone deliveries in rural areas since 2015 (Shen and Sun, 2023). Furthermore, in Europe, the European Union's Single European Sky ATM Research (SESAR) program has been exploring the integration of drones into the European airspace for various applications, including parcel delivery (Doole et al., 2020). These examples underscore the growing global interest in UAV-assisted delivery systems and highlight the importance of developing models and methodologies that can be applied to different geographical contexts and regulatory environments. However, the UAV's constraints, such as battery life and maximum payload, continue to limit its operational range. These limitations notwithstanding, it is important to point out that the truck and UAV are quite complementary to each other. First, whereas the truck is relatively slow and can easily handle many heavy packages, the UAV is faster and is particularly useful for the delivery of small, light packages. Consequently, the truck has a wider range of operation than the UAV. The well-studied pickup plus delivery problem (PDP) is characterized by a scenario where each vehicle, starting from a central location (usually a depot), picks up items from a certain location and drops them off at another. The goal is to determine vehicle routes for these pick-ups and deliveries while minimizing operational costs.

Consider a delivery service that needs to visit a specific set of customers. Additionally, suppose there is a requirement that some items must be picked up before they are delivered, as shown in Fig. 2(a). It follows that a possible solution to this pickup and delivery problem is depicted in Fig. 2(b).

According to Cordeau et al. (2007), there are three types of pickup and delivery problems. The first case is the single-commodity scenario, where each node involves the pickup or delivery of a singular type of good. The second is the two-commodity PDP, where each node can serve as both a pickup and delivery node. Finally, the case where each commodity or item corresponds to a unique pickup and delivery node is referred to as the n-commodity problem. Examples of this scenario are the "dial-a-ride problem" (DARP), where passengers are required to be transported from a starting point to a predetermined destination, and the urban courier service problem, where small packages or letters are transported from building to building.

The dial-a-ride and urban courier service problems are collectively referred to as the "vehicle routing problem with pickup and delivery" (VRPPD). This paper aims to extend the VRPPD to allow for UAV deliveries to eligible customers and to investigate the effects this has on cost while meeting customer demand. We refer to the resultant model as the Drone-Assisted Pickup and Delivery Problem (DAPDP).

This work builds on Murray and Chu (2015) in which a truck departs a depot with parcels and a UAV. As the truck makes deliveries, there is an opportunity to launch the UAV to make deliveries to customers in proximity to the truck's current location. The truck then continues on its route, making deliveries along the way, and subsequently collects the UAV at a customer site different from the original launch position. We extend this concept in several ways:

1. We provide a formal specification for multiple trucks working with multiple drones in the "drone-assisted pickup and delivery problem" (DAPDP) through a mixed-integer linear program (MILP). We restrict package pickup to trucks only because we believe safety concerns will limit the use of drones to deliveries only. The adoption of any new technology is safety-sensitive, and thus, any customer injury related to handling the drone could set back adoption for several years. For this reason, in our model, we restrict deployment and retrieval to the truck driver. Additionally, we prioritize cost over route length because we believe it informs practical business decisions. This mixed integer linear program (MILP) is presented in Section 3.2.
2. To boost the performance of MILP solvers when tackling DAPDP instances, we devise several sets of valid inequalities. We demonstrate that the valid inequalities we propose significantly enhance the solver's capability to find optimal solutions for smaller instances.
3. However, the application of the MILP formulation for determining optimal solutions within a tolerable time duration remains constrained to smaller instances. Hence, to tackle larger instances of the DAPDP, we develop a dynamic route and reassign heuristic in Section 4 that allows for the treatment of route optimizations as a whole and not as independent optimizations.
4. In any event, our numerical study demonstrates that employing drones can substantially improve the objective, thereby rendering them a promising option for last-mile delivery. Additionally, we offer an exhaustive sensitivity analysis of important UAV parameters.

This paper is structured as follows: Section 2 presents a review of the relevant literature on drone delivery. This literature review starts from studies involving single trucks and then progressively explores more complex configurations. The problem is then described explicitly and a mathematical model is formulated in Section 3. We formulate the model as a mixed integer linear program (MILP) based on the traveling salesman problem (TSP). A heuristic to solve problems of practical size is introduced and described in Section 4. The numerical results obtained by conducting various experiments are presented in Section 5 and finally, our conclusions are presented in Section 6.

2. Literature review

Progress in research and development has allowed UAVs to be utilized in a myriad of applications, which include the delivery of goods. Many applications extend the traditional traveling salesman problem (TSP), vehicle routing problem (VRP), and pickup and delivery problem (PDP) to include drone-assisted delivery, which is also the subject of this paper. The research pertaining to TSP, VRP, and PDP is substantial and well-documented, as detailed in Applegate et al. (2006), Toth and Vigo (2014), Golden et al. (2008), Eksioglu et al. (2009), Braekers et al. (2016) and Cordeau et al. (2007). We know that the TSP is generalized by the VRP, which in turn is generalized by the PDP. Among the variants of these problems, a number of studies consider the collaboration between trucks and UAVs for delivery. It follows that the TSP, now including drone, is generalized by the VRP with drone, which is generalized by the PDP with drone. This literature review also follows this categorization, first highlighting studies concerned with coordinating drone(s) with a single vehicle (TSP plus drone), then showcasing studies dealing with multiple vehicles with multiple drones (VRP plus drone). Finally, we detail studies dealing with the additional constraints of pickup and delivery (PDP plus drone). This paper addresses the operational challenges concerned with the latter category, i.e., PDP with drone.

The model proposed in this paper bears resemblance to the FSTSP formulation presented by Murray and Chu (2015), where the use of UAVs for last-mile delivery is investigated due to their complementary attributes with respect to trucks. Murray and Chu (2015) formulate an optimization problem focused on determining the best routing and scheduling for UAVs and trucks in the parcel delivery process. In their formulation, while the UAV completes its sortie, the truck is allowed to deliver to any number of customers, given that the UAV's battery life can accommodate a rendezvous with the truck at a customer location different from the launch site. Additionally, they introduce a heuristic solution methodology that utilizes the Clarke–Wright savings heuristic (Clarke and Wright, 1964). They also examine the sensitivity of the mathematical model to the drone parameters, namely speed and endurance.

Ponza (2016) extends the FSTSP formulation by simplifying the formulation of some constraints in the original FSTSP model. Ponza (2016) also presents a simulated annealing solution approach to the resulting mixed integer linear program. Further, numerical experiments demonstrate an improvement over truck-only deliveries. Ha et al. (2018) explore a modification of the FSTSP, where the primary objective shifts from makespan minimization, as in the original FSTSP framework, to cost reduction. This cost function includes a transportation cost component and one for the time wasted when one vehicle waits for the other. They also propose two solution methods, the first based on that of Murray and Chu (2015) and the second on a new split procedure that improves TSP solutions via drone delivery insertions.

Crişan and Nechita (2019) introduce an improved greedy heuristic based on that of Murray and Chu (2015) in which they also minimize route length, as in Murray and Chu (2015). However, they also propose a novel cost function that incorporates the flight duration of the drone. Starting from an initial optimal TSP solution for the truck, their approach maintains the node order as established in the initial solution, which reduces algorithmic complexity. Further, since drone savings restrict the drone to only fly between neighboring nodes, as in the initial solution, their approach is less computationally intensive. In his thesis, Yoon (2018) extends the model of Murray and Chu (2015) by considering the scenario of multiple drones working in tandem with each truck to minimize total cost. Similarly, Sacramento et al. (2019) extend the model of Murray and Chu (2015) by incorporating drones into the vehicle routing problem for capacitated trucks. They also propose a solution method based on an ''adaptive large neighborhood search'' (ALNS) strategy.

A different model of the TSP plus drone (denoted ''TSP-D'') is proposed in Agatz et al. (2018). The main difference in their model with respect to the FSTSP is that the UAV traverses the same road network as the truck. The assumption extends to assuming that the UAV's speed exceeds that of the truck by a multiplier denoted as α, making it possible to determine the maximum gain the truck–drone tandem provides in comparison to the truck-only case. Moreover, they devise a series of heuristics that follow a 'route-first, cluster-second' approach, utilizing methods from local search and dynamic programming.

Bouman et al. (2018) describe exact solution approaches for the TSP-D and demonstrate that their application of dynamic programming techniques handles larger problem instances more effectively compared to traditional mathematical programming methods. Wang et al. (2017) generalize the TSP-D to the ''vehicle routing problem with drones'' (VRP-D), in which a fleet of trucks works in tandem with drones to deliver goods. This Vehicle Routing Problem with Drones (VRP-D) considers a fleet of m uniform trucks, each outfitted with k drones, with the capability to either deliver packages directly or transport the drones within a comfortable range for them to complete the delivery. The VRP-D experiments focused on determining the maximum savings obtainable from truck–drone collaboration. The VRP-D was subsequently extended by Poikonen et al. (2017) to explicitly allow for a limited battery life and an allowance for cost in the objective function.

A few papers address the pickup-delivery problem with respect to coordination between trucks and drones. The first is Lin (2011) in which coordination between foot couriers and trucks is proposed. Our work is distinct in that the UAV can only serve one customer per sortie, as opposed to the foot couriers in Lin (2011). We also assume that the UAV travels faster and along the Euclidean edge in comparison to a foot courier, who is slower and limited to a great extent by the road network. The second is Karak and Abdelghany (2019) in which a mother-ship system in a 'one-to-many-to-one' problem is investigated. The vehicle, along with the drones, originate from a singular depot, tasked with the mission of delivering goods to customers, retrieving goods from these customers, and subsequently transporting these items back to the originating depot. With respect to Karak and Abdelghany (2019), our study allows for only one drone per truck and limits the number of deliveries to one per sortie.

Recent studies have highlighted the potential of UAVs to improve the efficiency of last-mile delivery and reduce environmental impacts. For example, Baldisseri et al. (2022) investigate the environmental benefits of drone delivery in urban environments, while Pinto and Lagorio (2022) focus on the gains in operational efficiency from integrating UAVs into existing logistics networks. With the rapid technological advancements in UAVs, various sectors are exploring their potential applications, with transport and logistics being at the forefront. As described in Merkert and Bushell (2020), the management and strategic integration of drones present both opportunities and challenges, especially in monitoring/inspection/data collection, photography, recreation and logistics. Their review offers insights into the current usage patterns of airborne drones and suggests future strategic directions for their effective usage, underscoring the necessity of our research in this domain.

In terms of methodology, recent advances in solution techniques have also been made. Nguyen et al. (2022) propose a metaheuristic algorithm for solving the vehicle routing problem with drones, while Zhang et al. (2022) explore machine learning-based approaches to optimize routing decisions in UAV-assisted delivery systems. Furthermore, Luo et al. (2021) extend the FSTSP model to consider multi-objective optimization, incorporating both cost and time objectives, while Baek et al. (2020) focus on energy-efficient routing and scheduling of UAVs in a truck–drone collaborative delivery context. Colajanni et al. (2023) developed a comprehensive optimization model for supply chain networks, emphasizing profit maximization through the combined use of trucks and drones. This model intricately incorporates the operational characteristics of drones, specifically addressing their inherent limitations like battery life and delivery range, within the framework of a constrained nonlinear optimization challenge conceptualized as a variational inequality problem.

Kyriakakis et al. (2023) introduce the Energy Minimizing Drone Routing Problem with Pickups and Deliveries (EM-DRP-PD), with the primary objective of reducing total operational energy expenditure. This model uniquely considers the impact of cargo weight on drone range capabilities. To address this problem, they have devised and applied three variants of the Greedy Randomized Adaptive Search Procedure metaheuristic algorithm. Gao et al. (2023) embark on an examination of optimizing the schedules for groups of trucks, each capable of transporting multiple drones. They formulate a mixed-integer programming model, and a column generation-based heuristic algorithm is designed to handle a larger number of customers. Lastly, Yin et al. (2023) explore truck–drone delivery with the additional constraint of time windows. A branch-and-price-and-cut algorithm, augmented by a bounded bidirectional labeling method, specifically designed to effectively manage the pricing sub-problem, is employed as their solution strategy.

In light of these recent developments, our study contributes to the literature by specifically addressing the pickup and delivery problem in a drone-assisted context. Our proposed model and solution methodology build upon the FSTSP framework and recent advances in solution techniques to provide a comprehensive approach to optimizing drone-assisted pickup and delivery operations.

In the following section, the ''drone-assisted pickup and delivery problem'' (DAPDP), a variant of the ''vehicle routing problem with pickup and delivery'' (VRPPD), is fully described and then formulated as a MILP.

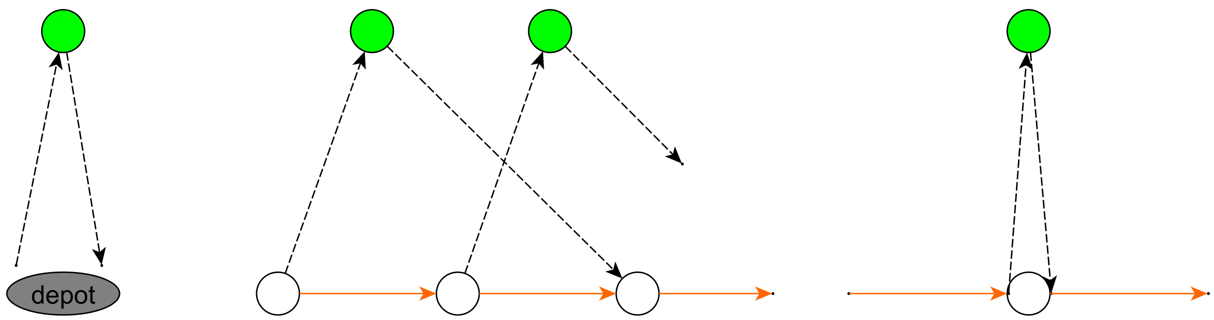


Fig. 3. UAV restrictions (UAV shown visiting green nodes while the truck visits the rest). Left: UAV cannot work independently of truck. Middle: Cannot launch a new sortie before the UAV has been recovered. Right: Launch and rendezvous points cannot be the same.

3. Problem description and formulation

We explicitly define the "drone-assisted pickup and delivery problem" (DAPDP) in this section, introduce the notation that will be used throughout the paper, and provide a mathematical formulation for the problem.

3.1. Problem description

The DAPDP considers a set of n pickup service requests with corresponding delivery nodes. Every request must be honored exactly once, with service provided by either a delivery truck or a UAV, which collaborates with the truck. Due to some UAV limitations, such as a payload limit or signature requirement, some customers can only be served by the truck.

The truck, along with the accompanying UAV, is required to set out from and return to a single depot once. Both vehicles start their journey together, but their return to the depot can either be in unison or independently. When not actively in service, the UAV is transported via truck, thereby ensuring conservative use of its battery life.

During a delivery cycle, the UAV is allowed to make multiple sorties, each involving three distinct locations. A sortie starts at a customer location, where it is loaded for delivery by the truck driver. A service time to load the package and change the battery, if necessary, is also accounted for. The next node in the sortie has to be UAV-eligible, and its corresponding pickup node must have previously been visited by the truck. The sortie's last node can be either the end depot or a node in the truck's route (but it cannot be the same as the launch node). Once launched, the UAV must complete the sortie before its battery drains out. After delivery, if the UAV flies to the truck's new location, a service time to retrieve the UAV is also accounted for. Neither the truck nor the UAV may revisit a customer. These restrictions on the movement of the UAV are depicted in Fig. 3. Finally, we note that the objective of the DAPDP is to minimize the overall cost associated with servicing all customer nodes via drone or truck.

3.2. Problem formulation

We present a mathematical formulation for the drone-assisted pickup and delivery problem (DAPDP) in this section. This formulation is an extension of the MILP proposed in Murray and Chu (2015) which incorporates UAV deliveries into routing problems based on the Traveling Salesman Problem (TSP).

Suppose we have a total of n pickup requests that need to be fulfilled. For each request i, we associate an originating node denoted as i and a destination node represented as $n+i$. If we assume that the trucks and UAVs are based at a single depot (dubbed nodes 0 and $2n+1$ for departures and arrivals, respectively), the DAPDP may be defined in the following sub-sections on a directed graph $\mathcal{G} = (\mathcal{N}, \mathcal{A})$, where $\mathcal{N}$ are the graph nodes (customer locations) and $\mathcal{A}$ are the directed arcs in the graph.

3.2.1. Sets

$\mathcal{C}$	Set of service nodes. $\mathcal{C} = \{1, 2, \ldots, n, n+1, \ldots 2n\}$
$\mathcal{C}'$	Subset of service nodes that the UAV may service. $\mathcal{C}' \subset \mathcal{C}$
$\mathcal{N}$	Set of nodes in the network (including depots). $\mathcal{N} = \{0, 1, \ldots, n, n+1, \ldots 2n+1\}$
$\mathcal{N}_0$	Set of all possible departure nodes for a vehicle. $\mathcal{N}_0 = \{0, 1, \ldots, n, n+1, \ldots 2n\}$
$\mathcal{N}_+$	Set of all nodes a vehicle can visit. $\mathcal{N}_+ = \{1, 2, \ldots, n, n+1, \ldots 2n+1\}$
$\mathcal{P}$	Set of tuples of the form $\langle i, j, k \rangle$ for 3-node sorties of UAV
ϕ^+	Set of all pickup nodes. $\phi^+ = \{1, 2, \ldots, n\}$
ϕ^-	Set of all delivery nodes. $\phi^- = \{n+1, n+2, \ldots, 2n\}$
$\mathcal{V}$	Set of homogeneous trucks. $\mathcal{V} = \{1, 2, \ldots, m\}$
$\mathcal{A}$	Set of feasible arcs. $\mathcal{A} = \{(i, j) : i \in \mathcal{N}_0, j \in \mathcal{N}_+\}$

3.2.2. Parameters

s_L	Time needed to load UAV before launch (minutes)
s_R	Time needed to recover UAV upon rendezvous (minutes)
e	Flight endurance of UAV (minutes)
s	Truck speed (miles per hour)
s'	Drone speed (miles per hour)
Q	Truck load capacity with drone (kg)
Q'	Drone load capacity (kg)
τ_{ij}	Time required by truck to move from node $i \in \mathcal{N}_0$ to node $j \in \mathcal{N}_+$ (minutes)
τ'_{ij}	Analogous time for UAV (minutes)
c_{ij}	Cost for truck to travel from node $i \in \mathcal{N}_0$ to $j \in \mathcal{N}_+$ ($)
c'_{ijk}	Cost to operate drone between nodes $i \in \mathcal{N}_0, j \in \mathcal{C}'$ and $k \in \mathcal{N}_+$ ($)
d_i	Truck service duration at node $i \in \mathcal{N}$ (minutes)
d'_i	Drone service duration at node $i \in \mathcal{C}'$ (minutes)

3.2.3. Decision variables

$$x_{ij}^v = \begin{cases} 1 & \text{if truck } v \in \mathcal{V} \text{ moves between nodes } i \in \mathcal{N}_0 \text{ and } j \in \mathcal{N}_+ \\ & \text{where } i \neq j. \\ 0 & \text{otherwise} \end{cases}$$

$$y_{ijk}^v = \begin{cases} 1 & \text{if UAV of truck } v \in \mathcal{V} \text{ departs from node } i \in \mathcal{N}_0, \text{flies to } j \in \mathcal{C} \\ & \text{and returns to the end depot or truck at node } k \in \mathcal{N}_+, \langle 0, j, k \rangle \in \mathcal{P}. \\ 0 & \text{otherwise} \end{cases}$$

$$p_{ij}^v = \begin{cases} 1 & \text{if a customer } i \in \mathcal{C} \text{ is visited at any given point before} \\ & j \in \{\mathcal{C} : j \neq i\} \text{ in the route of truck } v \in \mathcal{V}. \\ 0 & \text{otherwise} \end{cases}$$

3.2.4. Other variables

t_j^v	Time when truck $v \in \mathcal{V}$ reaches node $j \in \mathcal{N}_+$ (minutes)
$t_j^{v'}$	Time when UAV of truck $v \in \mathcal{V}$ reaches node $j \in \mathcal{N}_+$ (minutes)
t_0^v	Earliest time truck $v \in \mathcal{V}$ may leave depot (minutes)
$t_0^{v'}$	Earliest time UAV of truck $v \in \mathcal{V}$ may leave depot (minutes)
u_i^v	Specifies the position in truck $v \in \mathcal{V}$'s route of node $i \in \mathcal{N}_+$. $u_i^v \in \{1, 2, \dots, 2n+1\}$
w_i^v	Weight (load) of truck $v \in \mathcal{V}$ after visiting node $i \in \mathcal{N}$ (kg)

3.2.5. Mathematical model

$$\min \sum_{v \in \mathcal{V}} \left(\sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}^v + \sum_{i \in \mathcal{N}_0} \sum_{j \in \mathcal{C}} \sum_{k \in \mathcal{N}_+} (c'_{ij} + c'_{jk}) y_{ijk}^v \right) \tag{1}$$

$$\sum_{v \in \mathcal{V}} \left(\sum_{\substack{i \in \mathcal{N}_0 \\ i \neq j}} x_{ij}^v + \sum_{\substack{i \in \mathcal{N}_0 \\ i \neq j}} \sum_{k \in \mathcal{N}_+} y_{ijk}^v \right) = 1 \qquad \forall j \in \mathcal{C} \tag{2}$$

$$\sum_{j \in \mathcal{N}_+} x_{0j}^v \leq 1 \qquad \forall v \in \mathcal{V} \tag{3}$$

$$\sum_{i \in \mathcal{N}_0} x_{i,2n+1}^v \leq 1 \qquad \forall v \in \mathcal{V} \tag{4}$$

$$1 + u_i^v - u_j^v \leq (1 - x_{ij}^v)(2n+2) \qquad \forall i \in \mathcal{C}, j \in \{\mathcal{N}_+ : j \neq i\}, v \in \mathcal{V} \tag{5}$$

$$\sum_{\substack{i \in \mathcal{N}_0 \\ i \neq j}} x_{ij}^v = \sum_{\substack{k \in \mathcal{N}_+ \\ k \neq j}} x_{jk}^v \qquad \forall j \in \mathcal{C}, v \in \mathcal{V} \tag{6}$$

$$\sum_{\substack{j \in \mathcal{C} \\ j \neq i}} \sum_{k \in \mathcal{N}_+} y_{ijk}^v \leq 1 \qquad \forall i \in \mathcal{N}_0, v \in \mathcal{V} \tag{7}$$

$$\sum_{\substack{i \in \mathcal{N}_0 \\ i \neq k}} \sum_{j \in \mathcal{C}} y_{ijk}^v \leq 1 \qquad \forall k \in \mathcal{N}_+, v \in \mathcal{V} \tag{8}$$

$$2y^v_{ijk} \leq \sum_{\substack{h\in\mathcal{N}_0\\ h\neq i}} x^v_{hi} + \sum_{\substack{l\in C\\ l\neq k}} x^v_{lk} \qquad \forall i \in \mathcal{C}, j \in \{\mathcal{C} : j \neq i\}, k \in \mathcal{N}_+, v \in \mathcal{V} \tag{9}$$

$$y^v_{0jk} \leq \sum_{\substack{h\in\mathcal{N}_0\\ h\neq k}} x^v_{hk} \qquad \forall j \in \mathcal{C}, \forall k \in \{\mathcal{N}_+ :< 0, j, k >\in \mathcal{P}\}, v \in \mathcal{V} \tag{10}$$

$$u^v_k - u^v_i \geq 1 - (2n+2)\left(1 - \sum_{j\in C} y^v_{ijk}\right) \qquad \forall i \in \mathcal{C}, \forall k \in \{\mathcal{N}_+ : k \neq i\}, v \in \mathcal{V} \tag{11}$$

$$t^{v'}_i \geq t^v_i - M\left(1 - \sum_{\substack{j\in C\\ j\neq i}} \sum_{k\in\mathcal{N}_+} y^v_{ijk}\right) \qquad \forall i \in \mathcal{C}, v \in \mathcal{V} \tag{12}$$

$$t^{v'}_i \leq t^v_i + M\left(1 - \sum_{\substack{j\in C\\ j\neq i}} \sum_{k\in\mathcal{N}_+} y^v_{ijk}\right) \qquad \forall i \in \mathcal{C}, v \in \mathcal{V} \tag{13}$$

$$t^{v'}_k \geq t^v_k - M\left(1 - \sum_{\substack{i\in\mathcal{N}_0\\ i\neq k}} \sum_{j\in C} y^v_{ijk}\right) \qquad \forall k \in \mathcal{C}, v \in \mathcal{V} \tag{14}$$

$$t^{v'}_k \leq t^v_k + M\left(1 - \sum_{\substack{i\in\mathcal{N}_0\\ i\neq k}} \sum_{j\in C} y^v_{ijk}\right) \qquad \forall k \in \mathcal{C}, v \in \mathcal{V} \tag{15}$$

$$\begin{aligned} t^v_k \geq\ & t^v_h + \tau_{hk} + d_h + s_L\left(\sum_{\substack{l\in C\\ l\neq k}} \sum_{m\in\mathcal{N}_+} y^v_{klm}\right) + s_R\left(\sum_{\substack{i\in\mathcal{N}_0\\ i\neq k}} \sum_{j\in C} y^v_{ijk}\right) - M\left(1 - x^v_{hk}\right), \\ & \forall h \in \mathcal{N}_0, k \in \{\mathcal{N}_+ : k \neq h\}, v \in \mathcal{V} \end{aligned} \tag{16}$$

$$t^{v'}_j \geq t^{v'}_i + \tau'_{ij} - M\left(1 - \sum_{k\in\mathcal{N}_+} y^v_{ijk}\right) \qquad \forall j \in \mathcal{C}', i \in \{\mathcal{N}_0 : i \neq j\}, v \in \mathcal{V} \tag{17}$$

$$t^{v'}_k \geq t^{v'}_j + \tau'_{jk} + d'_j + s_R - M\left(1 - \sum_{i\in\mathcal{N}_0} y^v_{ijk}\right) \qquad \forall j \in \mathcal{C}', \forall k \in \{\mathcal{N}_+ : k \neq j\}, v \in \mathcal{V} \tag{18}$$

$$t^{v'}_k - (t^{v'}_j - \tau'_{ij}) \leq e + M\left(1 - y^v_{ijk}\right) \qquad \forall k \in \mathcal{N}_+, j \in \{\mathcal{C} : j \neq k\}, v \in \mathcal{V} \tag{19}$$

$$u^v_i - u^v_j \geq 1 - (2n+2)p^v_{ij} \qquad \forall i \in \mathcal{C}, j \in \{\mathcal{C} : j \neq i\}, v \in \mathcal{V} \tag{20}$$

$$u^v_i - u^v_j \leq -1 + (1 - p^v_{ij})(2n+2) \qquad \forall i \in \mathcal{C}, j \in \{\mathcal{C} : j \neq i\}, v \in \mathcal{V} \tag{21}$$

$$p^v_{ij} + p^v_{ji} = 1 \qquad \forall i \in \mathcal{C}, j \in \{\mathcal{C} : j \neq i\}, v \in \mathcal{V} \tag{22}$$

$$\begin{aligned} t^{v'}_l \geq\ & t^{v'}_k - M\left(3 - \sum_{\substack{j\in C\\ j\neq l}} y^v_{ijk} - \sum_{\substack{m\in C\\ m\neq i\\ m\neq k\\ m\neq l}} \sum_{\substack{n\in\mathcal{N}_+\\ n\neq i\\ n\neq k}} y^v_{lmn} - p^v_{il}\right) \\ & \forall i \in \mathcal{N}_0, k \in \{\mathcal{N}_+ : k \neq i\}, l \in \{\mathcal{C} : l \neq i, l \neq k\}, v \in \mathcal{V} \end{aligned} \tag{23}$$

$$x^v_{0,2n+1} = 0 \qquad \forall v \in \mathcal{V} \tag{24}$$

$$\sum_{\substack{j\in\mathcal{N}_+\\ i\neq j}} x^v_{ij} = \sum_{\substack{j\in\mathcal{N}_+\\ i\neq j}} x^v_{n+i,j} + \sum_{\substack{l\in\mathcal{N}_0\\ l\neq k}} \sum_{k\in\mathcal{N}_+} y^v_{l,n+i,k} \qquad \forall i \in \phi^+, v \in \mathcal{V} \tag{25}$$

$$u^v_i - u^v_{j-n} \geq 1 - (2n+2)\left(1 - \sum_{j\in C'} y^v_{ijk}\right) \qquad \forall i \in \mathcal{C}, j \in \phi^-, k \in \{\mathcal{N}_+ : k \neq i\}, v \in \mathcal{V} \tag{26}$$

$$t^v_{n+i} \geq t^v_i + \tau_{i,n+i} + d_i \qquad \forall i \in \phi^+, v \in \mathcal{V} \tag{27}$$

$$w_j^v \geq w_i^v + q_j + q_m y_{jmk}^v + M(1 - x_{ij}^v) \qquad \forall i \in \mathcal{N}_0, j \in \{\mathcal{N}_+ : i \neq j\}, m \in \mathcal{C}', k \in \mathcal{N}_+, v \in \mathcal{V} \tag{28}$$

$$t_i^v \geq 0 \qquad \forall i \in \mathcal{N}, v \in \mathcal{V} \tag{29}$$

$$t_i^{v'} \geq 0 \qquad \forall i \in \mathcal{N}, v \in \mathcal{V} \tag{30}$$

$$t_0^v = 0 \qquad \forall v \in \mathcal{V} \tag{31}$$

$$t_0^{v'} = 0 \qquad \forall v \in \mathcal{V} \tag{32}$$

$$t_{2n+1}^v \leq T_{max} \qquad \forall v \in \mathcal{V} \tag{33}$$

$$t_{2n+1}^{v'} \leq T_{max} \qquad \forall v \in \mathcal{V} \tag{34}$$

$$p_{0j}^v = 1 \qquad \forall j \in \mathcal{N}_+, v \in \mathcal{V} \tag{35}$$

$$1 \leq u_i^v \leq 2n+2 \qquad \forall i \in \mathcal{N}_+, v \in \mathcal{V} \tag{36}$$

$$0 \leq w_i^v \leq Q \qquad \forall i \in \mathcal{C}, v \in \mathcal{V} \tag{37}$$

We aim to minimize the overall cost incurred by the trucks and UAVs, which is proportional to the distances traveled, as stated in the objective (1). Constraint (2) mandates that each customer is served only once, while constraints (3) and (4) dictate that all trucks must depart from and return to the same depot at most once. Constraint (5) serves to prevent sub-tours for each truck, and constraint (6) imposes the condition that a truck that visits node j must also leave from the same node. Constraint (7) sets the condition that a UAV can take off from any specific node only once, while constraint (8) ensures that a UAV can only return to a given node at most once. Constraint (9) requires a truck to accompany a UAV to both its launch and rendezvous nodes, which means that if the UAV starts its sortie from customer node i and is picked up by the truck driver at node k, then the truck must visit both nodes i and k. Constraint (10) extends this principle to the starting node, so that if the UAV starts its sortie from the starting node 0 and the truck collects it at node k, then node k must also be part of the truck's route. Constraint (11) stipulates that if a UAV initiates its sortie from customer node i and the truck retrieves it at node k, the truck must visit node i prior to reaching node k.

The set of constraints (12) to (15) maintain temporal consistency. Constraints (12) and (13) synchronize the time of arrival of the truck and its accompanying UAV at customer node i before the UAV launches, while constraints (14) and (15) coordinate arrival times for the truck and UAV at customer node k upon UAV return. It is worth mentioning that these constraints permit different times of arrival for the truck and UAV at the end depot.

Constraint (16) accounts for the effect of UAV launch and recovery times on the effective travel duration of each truck between two nodes. For example, consider a truck traveling from node h along the arc to node k. The effective travel time between these two nodes takes into consideration the arrival time of the truck at node h, the time required to travel between nodes h and k, the package loading time in the case of a UAV launch from node k, and the UAV retrieval time, if necessary, at node k. Constraint (17) stipulates that if a UAV departs from node i, its arrival time at node j must include the travel time between nodes i and j. Additionally, if the UAV is retrieved at node k, its arrival time should factor in the travel time from node j to node k and the recovery time at node k, as defined by constraint (18) (Murray and Chu (2015)).

The restriction on the UAV's flight endurance is enforced by constraint (19). It ensures that the time between the UAV's return to node k and launch from node i does not exceed the UAV's flight endurance. Constraints (20) to (22) define the right values of p_{ij}^v, with u_i^v and p_{ij}^v specifying the sequence in which each truck visits nodes.

Constraint (23) is designed to eliminate the possibility of a new UAV launch during flight. This constraint states that if a UAV sortie starts at node i, makes a delivery, returns to node k, and then launches from node l (so that $p_{il}^v = 1$), the launch time from node l must not precede the return time to node k. As stated in constraint (24), truck movement between depots is prohibited. Each pickup and its corresponding delivery node have to be served by the same vehicle or its accompanying UAV, as indicated in constraint (25). Additionally, if a UAV sortie starts from service node i for delivery at node j, the truck must have visited the corresponding pickup node j-n before visiting node i, where n represents the number of pickup (or delivery) requests, as stated in constraint (26). The trucks must also visit a pickup node before their corresponding delivery node, as outlined in constraint (27).

Constraint (28) helps maintain weight consistency in the solution by enforcing the requirement that the weight of the packages must not exceed the capacity of the truck. The bounds for arrival times at each node for each truck-UAV pair are set by constraints (29) and (30). The departure times from the starting depot are determined by constraints (31) and (32). The duration of each truck-UAV combination's shift is restricted by constraints (33) and (34). Constraint (35) ensures that the starting depot is the first node on the truck's path, while constraint (36) sets logical bounds on the binary variable u_i^v. The capacity limit of each truck is enforced by the constraint (37).

3.3. Valid inequalities

In an effort to improve the efficiency of MILP solvers when tackling the DAPDP, we develop a number of valid inequalities (VIs) that serve as cuts within the problem's solution space without eliminating any optimal solutions. As a result, incorporating these VIs could potentially reduce the computational time necessary to solve the DAPDP.

3.3.1. Cost bounds

In this section, we discuss the formulation of valid inequalities that we utilize to establish bounds on the objective. Although deriving these valid inequalities is relatively simple, their substantial influence on the solver's performance is evident, as they offer an enhanced relaxation to the MILP solver (details in Section 5).

Proposition 1. *The MILP (1)–(37) is equivalent to the MILP below:*

$$min\ \eta \tag{38}$$

$$\eta \ \geq \ \sum_{v\in\mathcal{V}}\left(\sum_{(i,j)\in\mathcal{A}} c_{ij}x^{v}_{ij} + \sum_{i\in\mathcal{N}_0}\sum_{j\in\mathcal{C}}\sum_{k\in\mathcal{N}_+}(c'_{ij}+c'_{jk})y^{v}_{ijk}\right) \tag{39}$$

$Constraints$ (2)–(37)

Proof. The variable η signifies the expense of using collaborating trucks and drones to satisfy customer demand. In a viable DAPDP solution, this expense is the total of the individual costs of the assigned truck-UAV pairs. Hence, the sum of all costs for each truck and drone serves as a valid lower bound on the objective:

$$\eta \ \geq \ \sum_{v\in\mathcal{V}}\left(\sum_{(i,j)\in\mathcal{A}} c_{ij}x^{v}_{ij} + \sum_{i\in\mathcal{N}_0}\sum_{j\in\mathcal{C}}\sum_{k\in\mathcal{N}_+}(c'_{ij}+c'_{jk})y^{v}_{ijk}\right) \tag{40}$$

This concludes the proof of Proposition 1. □

3.3.2. DAPDP symmetry

In this section, we show that the DAPDP exhibits symmetric properties with symmetric solutions when considering a symmetric graph. We leverage this characteristic to obtain a collection of symmetry-breaking cuts, leading us to propose the following proposition:

Proposition 2. *Suppose we have a solution for the MILP (1)–(37) consisting of components x^{v}_{ij} and y^{v}_{ijk}. Additionally, in the provided solution, let π^{v} represent the order in which customers are visited by each truck $v \in \mathcal{V}$. Consequently, the following statements are true:*

- *The VRPD exhibits some form of symmetry, meaning that for each π^{v}, there exists an alternative solution with an identical objective value that involves visiting customers in the reverse order of π^{v}.*
- *The symmetry inherent in the DAPDP can be broken by introducing the inequality:*

$$\sum_{i\in\mathcal{N}} u^{v}_{i}.x^{v}_{0,i} \ \leq \ \sum_{j\in\mathcal{N}} u^{v}_{j}.x^{v}_{j,2n+1}, \qquad \forall v \in \mathcal{V} \tag{41}$$

Proof. For every truck $v \in \mathcal{V}$, the decision variables x^{v}_{ij} are directly linked to an ordering of customers π^{v}, in truck v's path. Due to symmetry in the problem, we can add a constraint ensuring that a path $\pi^{v} = \{0, i, \dots, j, 2n+1\}$ (where $x^{v}_{0,i} = x^{v}_{j,2n+1} = 1$) and its partially reversed sequence $\pi^{v} = \{0, j, \dots, i, 2n+1\}$ are mutually exclusive. With this in mind, we propose the following set of static symmetry-breaking inequalities (Schermer et al. (2019)). These inequalities ensure that for any sequence $\pi^{v} = \{0, i, \dots, j, 2n+1\}$, $i \leq j$, the partially reversed sequence $\pi^{v} = \{0, j, \dots, i, 2n+1\}$ is not considered.

$$\sum_{i\in\mathcal{N}} u^{v}_{i}.x^{v}_{0,i} \ \leq \ \sum_{j\in\mathcal{N}} u^{v}_{j}.x^{v}_{j,2n+1}, \qquad \forall v \in \mathcal{V} \tag{42}$$

This concludes the proof of Proposition 2. □

3.3.3. Knapsack inequalities

We introduce knapsack inequalities that are related to the logical deployment of trucks and corresponding drones.

Proposition 3. *Suppose we define the sets $i \in \mathcal{N}_0, j \in \{\mathcal{C}' : i \neq j\}, k \in \mathcal{N}_+$. Given these set definitions, the inequalities below are valid for the mixed integer linear program (1)–(37):*

$$x^{v}_{ij} + \sum_{\substack{k\in\mathcal{N}_+\\ i\neq k}} y^{v}_{ijk} \ \leq \ 1, \qquad \forall v \in \mathcal{V}, i \in \mathcal{N}_0, j \in \mathcal{C}', i \neq j \tag{43}$$

$$x^{v}_{ij} + x^{v}_{jk} + x^{v}_{ik} + y^{v}_{ijk} \ \leq \ 2, \qquad \forall v \in \mathcal{V}, i \in \mathcal{N}_0, j \in \mathcal{C}', i \neq j, i \neq k, j \neq k \tag{44}$$

Proof. Consider a pair of two vertices $i \in \mathcal{N}_0$, and $j \in C'$ where $i \neq j$. Constraints (2) ensure that the in-degree of completed deliveries does not exceed one. As a result, j can be served from i at most once, either by a truck or a drone. This observation can be expressed through the following inequalities:

$$x_{ij}^{v} + \sum_{\substack{k \in \mathcal{N}_+ \\ i \neq k}} y_{ijk}^{v} \leq 1, \qquad \forall v \in \mathcal{V}, i \in \mathcal{N}_0, j \in C', i \neq j \tag{45}$$

Now, consider a triplet of vertices $i \in \mathcal{N}_0, j \in C'$ and $k \in \mathcal{N}_+$, with the caveat that all vertices are distinct. In this triplet, when the truck and drone are positioned at i, a maximum of two deliveries can occur to satisfy all demands. Starting at i, the truck can either serve j first and then k, or it can directly travel from i to k. In the second scenario, a drone is allowed to deliver to j. This can be formulated mathematically with the following inequalities:

$$x_{ij}^{v} + x_{jk}^{v} + x_{ik}^{v} + y_{ijk}^{v} \leq 2 \qquad \forall v \in \mathcal{V}, i \in \mathcal{N}_0, j \in C', i \neq j, i \neq k, j \neq k \tag{46}$$

This concludes the proof of Proposition 3. □

4. The proposed DAPDP heuristic

The computational time necessary to solve the MILP defined by Eqs. (1)–(37) grows exponentially as the number of customers increases, reaching several hours for only a handful of customers. To solve problems of practical size, therefore, we propose a heuristic based on the FSTSP (flying sidekick traveling salesman problem) heuristic in Murray and Chu (2015).

4.1. Algorithm 1: Main algorithm

The FSTSP heuristic is a route-and-reassign technique that first solves a TSP and then reassigns some UAV-eligible nodes to the UAV for delivery based on the savings criteria. The savings is a concept that measures the decrease in cost achieved when two smaller routes are combined into one larger route, as outlined by Clarke and Wright (1964). The proposed DAPDP heuristic extends this by considering multi-truck UAV routes, for which a pickup must precede a delivery and for which the weight of the packages has to be less than a specified truck capacity as the truck picks up and drops off packages along its route. Additionally, the truck route duration, which corresponds to a work shift, is capped. Algorithm 1 highlights the pseudo-code of the main function for the proposed heuristic.

The Pickup and Delivery Problem (PDP) is $\mathcal{NP}$-hard, as it is a generalization of the Vehicle Routing Problem (VRP). We utilize Google's OR.Tools (2019) (see Appendix B) to first solve this pickup and delivery problem (PDP) since it has one of the most efficient VRP solvers (Nazari et al., 2018) available. For this section, we let `solvePDP` denote a function that returns a solution to the PDP.

The `truckRoutes` object returned by the `solvePDP` function contains the truck routes. Each truck on these routes sets off from the depot, serves a subset of customers, and subsequently returns to the depot. As an example, in Fig. 4(a), 12 customers are served by two trucks, with each truck visiting a subset of the customers. If n denotes the total number of pickup requests, then each request at the origin node i corresponds to a destination node $n+i$. The single depot is denoted as nodes 0 and $2n+1$. For example, in the case of 12 customers, there are 6 pickup requests (nodes 1 to 6) with respective deliveries at nodes 7 to 12. We also note that only the delivery nodes are eligible for UAV service.

Additionally, the arrival times for each truck are returned by the `solvePDP` function in the variable `t`. These arrival times are used to calculate the savings obtained by having the UAV serve some customer nodes instead of the truck. The heuristic, specified in line 3 of Algorithm 1, operates by iteratively examining each of the routes in `truckRoutes`. Consider one of the truck routes in Fig. 4(a). Initially, since no sub-routes are served by the UAV, the `truckRoutes` and `truckSubRoutes` are the same. Thus, `truckRoutes` is copied into `truckSubRoutes` before the procedure builds sub-routes, some of which will be served by the UAV. To track these sub-routes served by the UAV, we also include the variable `truckSubRoutesWithUAV` which is initially empty. This is illustrated in Fig. 4(b) for one of the truck routes in Fig. 4(a).

The heuristic proceeds in line 12 of Algorithm 1 by considering each UAV-eligible customer (`j ∈ Cprime`). If this UAV-eligible customer is served by the truck under consideration (`j ∈ truckRoute`), the position of each customer in the truck's route is determined and saved in `truckPositions`. For the truck route under consideration in Fig. 4(b), these positions are indicated in parentheses. These truck positions are used to ensure that a delivery is never made before a pickup. For example, delivery to customer 8 ∈ `Cprime` (position 4) cannot be made before its corresponding pickup at customer 2 (position 1).

4.2. Algorithm 2: Calculate maximum displacement

Next, the heuristic in line 15 of Algorithm 1 calls the `calcMaxDisplacement` function in Algorithm 2. To illustrate the utility of this function, consider Figs. 4(c) and 4(d). As the truck visits customers, its weight fluctuates as it drops off and picks up packages. This is illustrated in Fig. 4(c) and the corresponding array is determined in lines 1–4 in Algorithm 2. Now, we also know that each truck has a weight limit denoted by `vehCapacity` and we wish to determine the effect on this vehicle capacity of the insertion of a customer at another point in the truck's route. If we consider customer 8 ∈ `Cprime` in Fig. 4(c), we note that moving its delivery closer to its pickup (customer 2) would not have any effect on the capacity since the delivery would be earlier in the

Algorithm 1: DAPDP heuristic pseudo-code

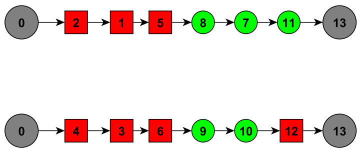
```
Require:
Ensure:
 1: Cprime ← C' {Copy of set of UAV-eligible customers}
 2: (truckRoutes, t) ↤ solvePDP(C)
 3: for all (truckRoute in truckRoutes) do
 4:   if at least one route has already been partitioned then
 5:     Call the performTopUp function
 6:   end if
 7:   truckSubRoutes ← [truckRoute]
 8:   truckSubRoutesWithUAV ← ∅
 9:   maxSavings ← 0
10:   maxSavingsFlag ← True
11:   while maxSavingsFlag ≠ False do
12:     for all (j ∈ Cprime) do
13:       if j ∈ truckRoute then
14:         Determine position of each node in the truck's path. Save as truckPositions.
15:         Call the calcMaxDisplacement function
16:         Call the calcSavings function
17:         for each subroute in truckSubRoutes do
18:           if subroute in truckSubRoutesWithUAV then
19:             Call the calcCostTruck function
20:           else
21:             Call the calcCostUAV function
22:           end if
23:         end for
24:       end if
25:     end for
26:     if maxSavings are positive then
27:       Call the performUpdate function
28:       Reset the maxSavings to 0
29:     else
30:       Stop
31:     end if
32:   end while
33:   Save truckRoute and respective truckSubRoutes and truckSubRoutesWithUAV in truckRouteSolutions.
34: end for
Output: truckRouteSolutions
```

truck's route. However, if the delivery is moved farther along the truck's path than its current position, it is possible to exceed the vehicle's capacity. For delivery to customer 8, the worst-case scenario is shown in Fig. 4(d) where it has been displaced from its current position and moved towards the return depot. We note that the effect is that the truck departs the other nodes with an additional weight due to a now undelivered package destined for customer 8. We use this realization in lines 5–10 of Algorithm 2 by adding the weight of the delivery customer under consideration (`j`) to the truck's weight for nodes farther along the route than `j`. The `maxDisplacement` is then the first node for which this sum exceeds the vehicle capacity.

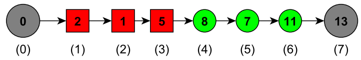
4.3. Algorithm 3: Calculate savings

In line 16, the algorithm 1 computes the savings gained by removing the UAV-eligible customer from its current position on the truck's path. These `savings` are calculated by calling the `calcSavings` function in Algorithm 3. If we consider moving customer 8 ∈ `Cprime` in Fig. 4(b), the predecessor and successor nodes would be `i`= 5 and `k`= 7, respectively, and the associated savings would be given by $\tau_{58} + \tau_{87} - \tau_{57}$. That is, the truck would travel directly on arc 5 → 7 rather than on arcs 5 → 8 → 7. The savings calculation is straightforward at this point because there is no UAV associated with this route as yet.

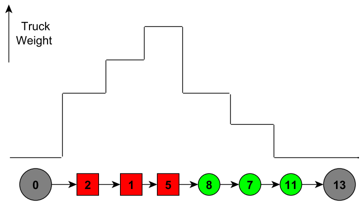
However, if a customer is on a subroute that is associated with a UAV, the available `savings` may be affected by the UAV sortie. This is because removing customer 8 from its location could result in the truck having to wait for the UAV to return, which would inject idle time into the truck's route. Considering Fig. 4(e), for customer `j` = 8, there is a UAV associated with subroute {5,8,7,13} serving a customer at `j'` = 11. The UAV flies from the first node in the subroute, `a` = 5, and is retrieved at the return depot, `b` = 13. In this case, the savings are subject to the UAV flight time on arcs 5 → 11 → 13, as calculated in line 10 of Algorithm 3. If the truck is required to wait for the UAV's return at the rendezvous node `b`, then the calculated `savings` will be negative. Otherwise, the UAV waits for the truck, and the `savings` are positive. These positive `savings` are preferable since they limit truck idle time.



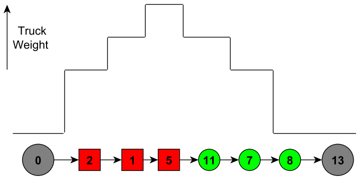
(a) Two trucks initially assigned to visit all customers. Green circular nodes represent UAV-eligible customers. Red square nodes represent UAV-ineligible customers



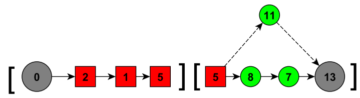
(b) For this truck, initially `truckSubRoutes = truckRoute =` {0,2,1,5,8,7,11,13}. Cprime = {7,8,11}. `truckSubRoutesWithUAV=` ∅



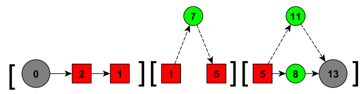
(c) Total weight of the truck as it makes deliveries. It should not exceed the truck capacity



(d) Different weight profile if order of deliveries is changed



(e) Drone makes delivery to customer at node 11. `truckRoute =` {0,2,1,5,7,8,13}



(f) Drone makes deliveries to customers at nodes 7 and 11

Fig. 4. Customers eligible for UAV delivery appear as green circular nodes, while UAV-ineligible customers are depicted as red square nodes. UAV deliveries are represented by dashed lines.

Algorithm 2: `calcMaxDisplacement`: Calculates the maximum position in the truck route that a UAV-eligible customer `j` can be re-assigned.

Require: `j, demands, vehCapacity, truckRoute, truckPositions`
Ensure:

```
{Determine truck's weight when it departs each node}
for all customer in truckRoute do
    Add weight of current node to truck's weight and save
end for
Set maxDisplacement to last position in truckRoute
for all nodes between truckPositions(j) and last do
    {Evaluate whether repositioning customer j further along the truck's route would result in
    the truck's weight exceeding its capacity, vehCapacity.}
    Add demand at customer j to the truck weight
    Note the first position at which the truck weight would exceed
    vehCapacity and save that position as maxDisplacement
end for
```

Output: `maxDisplacement`

4.4. Algorithm 4: Calculate truck cost

In the next step, as highlighted in line 17 of the main Algorithm 1, for a given customer `j`, the algorithm determines the cost of inserting this customer into another point in the truck's route (using the `calcCostTruck` function, as detailed in Algorithm 4), or alternatively, the cost of serving the same customer `j` via a UAV (utilizing the `calcCostUAV` function outlined in Algorithm 5). Considering customer $7 \in$ `Cprime` in Fig. 4(e), one option would be an insertion into the path of the truck in subroute {0,2,1,5}. On the other hand, customer $7 \in$ `Cprime` can be served by the UAV within the same subroute {0,2,1,5}. This is possible since the UAV already serves customer 7's current subroute, {6,8,7,13}.

Algorithm 3: `calcSavings`: Calculates the savings obtained when some UAV-eligible customer `j` is removed from the truck's route.

Require: `t, j, truckSubRoutesWithUAV`
Ensure:
Identify `i`, the node immediately before `j` in `truckRoute`
Identify `k`, the node immediately after `j` in `truckRoute`
Calculate: `savings` $\leftarrow \tau_{ij} + \tau_{jk} - \tau_{ik}$
if `j` $\in$ `subroute` $\in$ `truckSubRoutesWithUAV` **then**
  {`savings` might be affected by UAV `sortie`}
  Determine `a`, the first node in that truck `subroute` that includes `j`.
  Determine `b`, the last node in that truck `subroute` that includes `j`.
  Identify `j'`, the node that the UAV visits within this `subroute`.
  Determine `t'[b]`, the time of the truck's arrival at `b` if `j` is excluded from the truck's route.
  Calculate: `savings` $\leftarrow \min\{$`savings`$,\ t'[b] - (t[a] + \tau'_{a,j'} + \tau'_{j',b} + s_R)\}$
end if
Output: `savings`

Algorithm 4: `calcCostTruck`: Calculates cost of placing a customer `j` into a different point in the route of the truck.

Require: `t, j, subroute, maxDisplacement, savings, maxSavings, truckPositions`
Ensure:
1: Identify `a`, which is the start node in the truck's `subroute`.
2: Identify `b`, which is the end node in the truck's `subroute`.
3: Find `pickup`, the corresponding pickup node for customer `j`
4: **if** `truckPositions(b)` > `truckPositions(pickup)` **then**
5:   **for** (each pair of adjacent nodes, `i` and `k`, in the `subroute`) **do**
6:     {Attempt to insert customer `j` into this truck's `subroute`}
7:     **if** `j` $\neq$ `i` and `k` $\neq$ `j` **then**
8:       **if** `truckPositions(i)` $\geq$ `truckPositions(pickup)` **then**
9:         **if** `truckPositions(i)` < `maxDisplacement` **then**
10:           Calculate: `cost` $\leftarrow \tau_{ij} + \tau_{jk} - \tau_{ik}$
11:           **if** `savings` > `cost` **then**
12:             {Check that UAV can still fly the route, except if returning to depot}
13:             **if** `e` $\geq$ `t[b] - t[a] + cost` **then**
14:               **if** `maxSavings` < `savings - cost` **then**
15:                 {Save the change(s)}
16:                 `servedByTheUAV` $\leftarrow$ `False`
17:                 `(i*=i, j*=j, k*=k, servedTheByUAV)`
18:                 `maxSavings` $\leftarrow$ `savings - cost`
19:               **end if**
20:             **end if**
21:           **end if**
22:         **end if**
23:       **end if**
24:     **end if**
25:   **end for**
26: **end if**
Output: `(maxSavings, i*, j*, k*, servedTheByUAV)`

If the subroute under consideration is served by a UAV, then the main Algorithm 1 calls the `calcCostTruck` function in Algorithm 4. The function `calcCostTruck` calculates the cost associated with inserting customer `j` between two adjacent nodes, `i` and `k`. For customer `j` = 7 from Fig. 4(e), an insertion can be made between nodes 0 and 2, 2 and 1, and 1 and 5. Additionally, it ensures that customer `j` can only be inserted in the truck's path at a point after their corresponding pickup has been made. Further, it ensures that an insertion into the truck's path does not violate the truck capacity constraint. This cost of insertion is calculated in line 10 of Algorithm 4. To accept an insertion, Algorithm 4 first determines that the UAV can still feasibly fly the route since this insertion increases the truck's travel time. This constraint is ignored if the UAV is returning to the depot, since the trucks and the UAVs are permitted to return at different times. It then ensures that the net savings are greater than the `maxSavings` seen so far.

4.5. Algorithm 5: Calculate UAV cost

However, if the subroute under consideration is not served by the UAV, then the main Algorithm 1 calls the `calcCostUAV` function in Algorithm 5.

Algorithm 5: `calcCostUAV` determines the cost of serving a specific customer, `j`, via the UAV.

Require: `t, j, subroute, savings, maxSavings, truckPositions`
Ensure:
1: {Determine best possible UAV sortie for `j`}
2: Determine `t'`, the arrival time of the truck at all customer nodes, assuming `j` is removed from the truck's route.
3: Find `pickup`, the corresponding pickup node for delivery at customer `j`
4: **if** `truckPositions(b) > truckPositions(pickup)` **then**
5:   **for** (all pairs `(i, k)` within the `subroute`, where `i` precedes `k`) **do**
6:     **if** `j` ≠ `i` and `j` ≠ `k` **then**
7:       **if** `truckPositions(i)` ≥ `truckPositions(pickup)` **then**
8:         **if** $e \geq \tau'_{i,j} + \tau'_{j,k}$ **then**
9:           `cost` $\leftarrow \max\left\{\max\{(t'[k]-t[i]) + s_L + s_R,\ \tau'_{i,j} + \tau'_{j,k} + s_L + s_R\} - (t'[k]-t[i]),\ 0\right\}$
10:           **if** `maxSavings < savings - cost` **then**
11:             {Save the change}
12:             `servedByTheUAV ← True`
13:             `(i*=i, j*=j, k*=k, servedByTheUAV)`
14:             `maxSavings ← savings - cost`
15:           **end if**
16:         **end if**
17:       **end if**
18:     **end if**
19:   **end for**
20: **end if**
Output: `(maxSavings, i*, j*, k*, servedByTheUAV)`

The `calcCostUAV` function determines the cost of using the UAV to serve some customer `j`. For pairs of nodes where `i` precedes `k`, the function tries to find the most attractive launch and rendezvous locations (`i` and `k` respectively). For Fig. 4(e), if `j` = 7 were to be visited by the UAV in the first subroute, the (`i,k`) pairs to consider would be [(0,2), (0,1), (0,5)] and [(2,1), (2,5), (1,5)].

The `calcCostUAV` function ensures that the launch node for delivery to customer `j` does not precede the corresponding pickup node for customer `j` and that the UAV battery level is sufficient to fly the sortie $\langle i, j, k\rangle$. Given that customer `j` would be excluded from the truck's route, the algorithm recalculates the truck's arrival times at each node, represented as `t'`. The cost of serving customer `j` via the UAV is then calculated in line 9 of Algorithm 5. This cost is the maximum of either the additional time taken to launch and retrieve the UAV at node `i` or the time the truck has to wait for the UAV's return at rendezvous node `k`.

4.6. Algorithm 6: Perform update

At the end of each iteration (line 26 in Algorithm 1), an update corresponding to the maximum savings available is performed in Algorithm 6. If the customer is to be serviced by the UAV, then the update partitions that subroute to include a subroute that begins at the launch node and terminates at the rendezvous node. To allow the UAV to be launched or retrieved from these launch and rendezvous nodes, they are included as the last and first nodes in the adjacent subroutes, respectively. In Fig. 4(e), the first subroute {0,2,1,5} also includes the launch node because it can still be used as a rendezvous point for the UAV in that subroute. We also note that customer 7 ∈ `Cprime`, for example, can only be served by the UAV in the first subroute {0,2,1,5} since its current subroute {5,8,7,13} already has a UAV associated with it. Fig. 4(f) depicts this customer 7 ∈ `Cprime` being serviced by the UAV and the additional subroute partitioning that is performed.

On the other hand, if the customer corresponding to the maximum savings available is not to be served by the UAV, a simple insertion at the appropriate location is performed. Algorithm 1 then continues until no improvement in reassignments can be found.

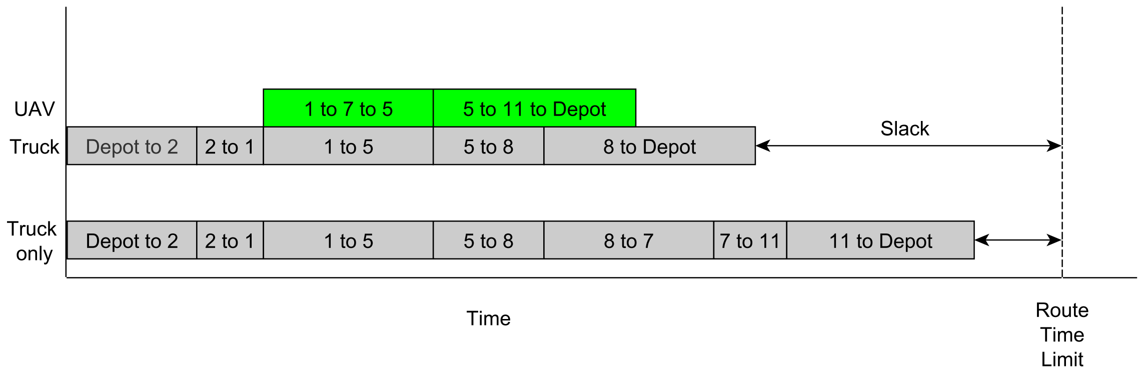
At this point, the algorithm has created a lot of slack in the route by shortening it. Fig. 5(a) illustrates a typical situation. However, since our objective is to reduce cost, Algorithm 1 proceeds in line 4 to utilize this slack by first allocating it to some of the nodes in subsequent routes before performing the UAV reassignments on those routes. From our example in Fig. 4, the slack available in Fig. 4(f) is first allocated to some nodes in the second route, i.e., route {0,4,3,6,9,10,12,13}. This slack is depicted in Fig. 5(a) which shows the space created with respect to the route time limit. Next, we discuss how this slack is optimally redistributed.

4.7. Algorithm 7: Perform top up

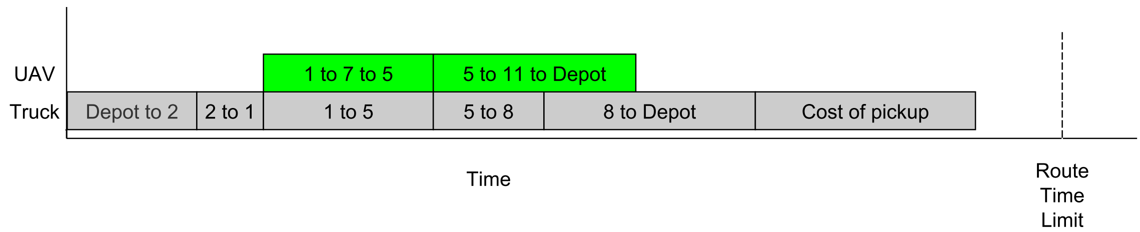
Algorithm 7 tries to allocate any slack in the current solutions saved in `truckRouteSolutions` to the nodes in the current route `truckRouteCurrent` with the goal of minimizing the number of trucks that have to depart the depot and thus the overall cost of operation. Since a delivery must be made after a corresponding pickup, the top-up operation is based on minimizing the cost of the pickup operation and then determining if it is feasible to perform this least-cost pickup via truck and the corresponding delivery via UAV or truck. If these twin operations are not feasible, a roll-back operation is performed. As such, the algorithm begins by determining the eligible set of pickup nodes and assigning them to `setPickupEligible`. The algorithm then initializes by assigning a suitably large number, denoted by `M`, to the minimum cost, `minCost`.

Algorithm 6: `performUpdate`: Updates the truck route and partitions the truck route into subroutes.

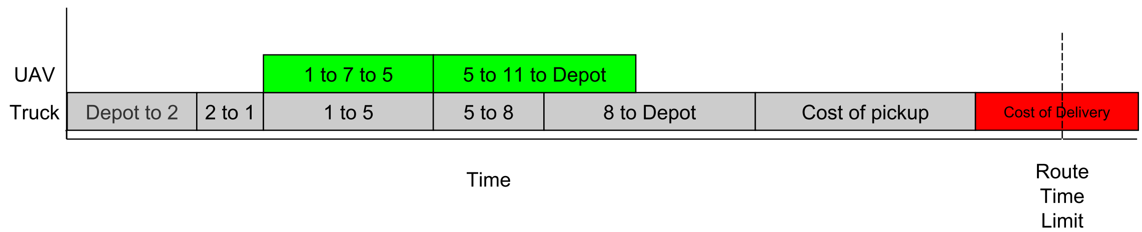
Require: `i*, j*, k*, servedByTheUAV`
Ensure:
if (the flag `servedByTheUAV` is True) **then**
  {Assign the UAV to sortie `i*-->j*-->.k*`}
  Delete j* from the `truckRoute`, `truckSubRoutes` and `truckSubRoutesWithUAV`
  Add a new truck subroute, starting from `i*` and ending at `k*`, to both `truckSubRoutes` and `truckSubRoutesWithUAV`.
  Delete the tuple (`i*, j*, k*`) from the set `Cprime`
  Update the arrival times of the truck at each node
else
  Delete `j*` from its position in the current truck subroute
  Insert `j*` between nodes `i*` and `k*` in the newly established truck subroute
  Update the truck route (`truckRoute`) and the truck's arrival times at each node
end if
Output: `(t, truckRoute, truckSubRoutes, truckSubRoutesWithUAV, Cprime)`



(a) Time slack introduced by algorithm



(b) Time cost due to pickup



(c) Time cost due to delivery

Fig. 5. The Top Up algorithm. Illustration of the time slack introduced and how it is utilized in the algorithm.

Algorithm 7: performTopUp: Add pickup and corresponding delivery to modified routes

```
Require: truckRouteSolutions, truckRouteCurrent
Ensure:
1: setPickupEligible ← [All pickup nodes ∩ truckRouteCurrent]
2: for all solutions in truckRouteSolutions) do
3:   minCost ← M
4:   minCostFlag ← True
5:   while minCostFlag ≠ False do
6:     Call the costTopUpPickup function
7:     Call the costTopUpDelivery function
8:   end while
9: end for
Output: truckRouteSolutions, truckRouteCurrent
```

4.7.1. *Algorithm 8: Add pickup node*

Algorithm 7 then calls the `costTopUpPickup` function of Algorithm 8 to determine which pickup node should be removed from the route of the current truck under consideration, `truckRouteCurrent`, and assigned to one of the other trucks. Since pickup nodes can only be served by trucks, the `costTopUpPickup` function only considers the cost of serving the pickup via `calcCostTruck` in line 4 of Algorithm 8. This least-cost pickup is then inserted in the route of the truck in line 8 if the additional time required to serve this pickup does not violate the time limit. This situation is depicted in Fig. 5(b).

Algorithm 8: costTopUpPickup: Add pickup node to modified routes

```
Require: setPickupEligible, minCost
Ensure:
1: for all i in setPickupEligible) do
2:   Call the calcMaxDisplacement function
3:   for all (subroute in truckSubRoutes) do
4:     Call the calcCostTruck function
5:   end for
6: end for
7: if route duration with pickup inserted would not exceed limit then
8:   Call the performUpdate function
9:   Reset minCost ← M
10:  if setPickupEligible is empty then
11:    minCostFlag ← False
12:  else
13:    Remove pickup* from setPickupEligible
14:  end if
15: else
16:  Reset minCost ← M
17: end if
Output: (minCost, minCostFlag, pickup*)
```

4.7.2. *Algorithm 9: Add corresponding delivery node*

Next, in line 7 of Algorithm 7, a call is made to the `costTopUpDelivery` function in Algorithm 9 to determine if the corresponding delivery can be carried out by the truck or its UAV. If the delivery is feasible (time limit is not violated), the delivery is inserted into the truck's route; otherwise (Fig. 5(c)), any changes are rolled back and control is returned to the main Algorithm 1.

4.8. *Theoretical computational complexity*

We wrap up this section by deriving the time complexity of the proposed DAPDP heuristic. The time complexity of an algorithm is a fundamental metric that quantifies the amount of time an algorithm takes to complete based on the length of its input.

If we let M denote the number of vehicles or truck routes and N denote the number of customers, the time complexity of the DAPDP heuristic summarized in Algorithm 1 is $O(M \cdot N^2)$. Step-wise:

- Loop 3 injects $O(M)$ time complexity.
- The call to function `performTopUp` in line 5 adds $O(N)$ time complexity.
- Loop 12 injects $O(N)$ time complexity.
- The call to function `calcMaxDisplacement` in line 15 adds $O(N)$ time complexity.
- The call to function `calcSavings` in line 16 adds $O(1)$ time complexity.

Algorithm 9: costTopUpDelivery: Add a corresponding delivery to modified routes

```
Require: pickup*
Ensure:
Determine delivery node that corresponds to pickup*
Call the calcMaxDisplacement function
for each (subroute in truckSubRoutes) do
  if subroute in truckSubRoutesWithUAV then
    Call the calcCostTruck function
  else
    Call the calcCostUAV function
  end if
  if route duration with delivery inserted would not exceed limit then
    Call the performUpdate function
    Remove delivery* node from truckRouteCurrent
    Remove corresponding pickup* node from truckRouteCurrent
  else
    Undo insertions in truckRoute, truckSubRoutes and truckSubRoutesWithUAV
    Update the time of arrival of the truck at all nodes
  end if
end for
Output: delivery*
```

- The call to function `calcCostTruck` in line 19 adds $O(N)$ time complexity.
- The call to function `calcCostUAV` in line 21 adds $O(N)$ time complexity.
- The call to function `performUpdate` in line 27 adds $O(N)$ time complexity.

5. Numerical experiments and discussion

In this section, a series of experiments are conducted to demonstrate the effectiveness of the proposed DAPDP heuristic. These tests also aim to investigate the trade-offs between the different parameters associated with drone utilization, with the intention of determining their relative importance in the performance of the heuristic. We begin in Section 5.1 by describing the parameters in the base case of the problem, and then discuss how the test instances used in the experiments are generated in Section 5.2. We carry out an evaluation of the heuristic's performance on small-scale and larger instances in Sections 5.2.1 and 5.2.3 respectively, and then perform a similar analysis on the closest benchmark instances in Section 5.3. Finally, a sensitivity analysis is carried out by adjusting the parameters of interest, as detailed in Section 5.5. This analysis will help to further understand the impact of each parameter on the overall solution.

All computational experiments were performed on a desktop PC, which was equipped with a quad-core Intel i7-7700 processor and 8 GB of RAM and running a 64-bit version of Windows 10. The mathematical model was written in GAMS (Bussieck and Meeraus, 2004) and the different scenarios were solved via the CPLEX solver. The algorithms for the heuristic in the experiments were implemented using the Python programming language, specifically version 3.6.5.

5.1. Parameters

The distance, d_{ij}, between nodes i and j is Euclidean, with the corresponding travel times $\tau_{i,j}, \tau'_{i,j}$ for truck and drone, respectively, being determined by the distance-speed-time relationship.

The speeds of the trucks and UAVs are set at 35 and 50 mph, respectively. In our model, we considered a delivery drone's top speed of 50 mph, reflecting the maximum achievable velocity under optimal conditions. This specification aligns with recent data about Amazon's delivery drone fleet, which, as per a CNET report, can achieve speeds up to 50 mph when delivering packages weighing up to 5 pounds (2.27 kg), while maintaining an altitude of 400 ft (Musil, 2022).

The truck operation cost, c_{ij}, as a function of distance, d_{ij}, fuel price, f_p, and fuel consumption rate, f_c, is given by the formula $c_{ij} = f_p \cdot f_c \cdot d_{ij}$ following Sacramento et al. (2019). The corresponding UAV cost, c'_{ij}, is assumed to be a fraction of the truck cost, $c'_{ij} = \alpha \cdot c_{ij}$ where α is designated the drone cost factor.

We anticipate that the maximum service duration for a truck at a node will be roughly twice that of the drone. These are chosen at random from the intervals [1,6] min and [1,3] min, respectively. These parameters are summarized in Table 1.

5.2. Random test instances

In order to evaluate the performance of our algorithm, we generated a set of random instances for testing. The lack of real-world datasets relevant to this particular problem domain made this approach necessary. In the generated instances, we randomly assign the coordinates for pickup and delivery nodes within a grid of size $d \times d$. This grid size d is sampled from a uniform distribution

Table 1
Summary of parameters used in the experiments.

Parameter	Notation	Value	Reference
Truck Speed	s	35 mph	Sacramento et al. (2019)
Drone Speed	s'	50 mph	Sacramento et al. (2019)
Launch Time	s_L	1 min	Murray and Chu (2015)
Recovery Time	s_R	1 min	Murray and Chu (2015)
Endurance	e	30 min	Murray and Chu (2015)
Truck Capacity with Drone	Q	1300 kg	Sacramento et al. (2019)
Truck Capacity without Drone	Q^*	1400 kg	Sacramento et al. (2019)
Drone Capacity	Q'	2.27 kg	Murray and Chu (2015)
Fuel Price	f_p	0.840 $/l	See text
Fuel Consumption	f_c	0.07 l/km	Sacramento et al. (2019)
Miles Converter	m_c	1.61 km/mi	Standard
Drone Cost Factor	α	10%	Sacramento et al. (2019)
Maximum Route Duration	T_{max}	8 h	See text
Service Duration for Truck	d_i	[1–6] min	See text
Service Duration for Drone	d'_i	[1–3] min	See text

$\mathcal{U}(0,d)$. The depot is then chosen to be the approximate center of gravity, i.e., the mean of the x- and y-coordinates of these pickup and delivery nodes. The instances generated through this process are labeled as "n.s.g", where n denotes the total number of customers within the instance, s signifies the grid size, and g is a generic name provided to the instance. For the purpose of these experiments, instances are generated with grid sizes ranging from 5 × 5 miles to 40 × 40 miles.

According to Amazon, up to 86% of the items they deliver are under 5 lb (2.27 kg) (Rose, 2013). Additionally, according to UPS (2013), individual packages cannot exceed 150 lb (68 kg). Following these constraints, the demands at the delivery locations are generated in line with a uniform distribution function, with the main consideration being whether or not the customer can be served by a drone. Let c_d denote a delivery customer in the instance, and let $0 \leq r < 1$ represent a randomly generated number associated with that customer. The demand at the delivery nodes, measured in kilograms, is determined as follows:

$$q(c_d, r) = \begin{cases} q_{c_d} \in \; -\mathcal{U}(0, 2.27) & \text{if } r < 0.86 \\ q_{c_d} \in \; -\mathcal{U}(2.27, 68) & \text{otherwise} \end{cases} \tag{47}$$

The corresponding demand at the pickup node c_p is represented by:

$$q(c_p, r) = -q(c_d, r) \tag{48}$$

5.2.1. Small-scale instances

In this section, the performance of the DAPDP heuristic is evaluated by comparing it to the optimal solution of the mixed integer linear programming (MILP) model. To evaluate the performance, small instances of the MILP are solved to optimality using the CPLEX solver in GAMS. The results from these optimal solutions are then compared with those obtained from solving the heuristic in Python.

The 48 test scenarios in this section are constructed with the number of customers (pickups and deliveries) in {6,8,10,12} randomly generated on grids of sizes 5 × 5, 10 × 10 and 20 × 20.

The optimality gap is calculated according to Eq. (49) for every instance, using z^{DAPDP}, the best objective found using the truck–drone approach, and z^*, the optimal solution of the MILP.

$$\frac{z^{DAPDP}}{z^*} - 1 \tag{49}$$

The results of these experiments are summarized in Fig. 6 with the full results in Table A.5 in Appendix A.

Table A.5 shows that the optimal solution z^{DAPDP} for the heuristic is reached relatively easily with an average execution time t^{DAPDP} of 0.008 s. The optimality gap also averages 6.4% with a standard error of 0.5%. These results prove the effectiveness of the heuristic. Additionally, Table A.5 illustrates that for every increase in the number of customers, the execution time for the MILP, t^{MIP}, grows exponentially.

5.2.2. Impact of valid inequalities

We now examine the effect of the valid inequalities introduced in Section 3.3 on the performance of the solver. We add the inequalities in turn to MILP (1)–(37). The results summarizing the time improvement in the performance of the solver due to the addition of the respective inequalities are shown in Table 2.

5.2.3. Large-scale instances

Since the PDP is $\mathcal{NP}$-hard, it follows that the DAPDP is also $\mathcal{NP}$-hard and therefore computationally expensive to solve for larger instances. This computation time becomes prohibitive for customer numbers greater than 10. In order to assess the quality of solutions produced by the DAPDP heuristic for large-scale instances, we propose comparing these solutions to a baseline scenario where only trucks are used for deliveries without any assistance from drones.

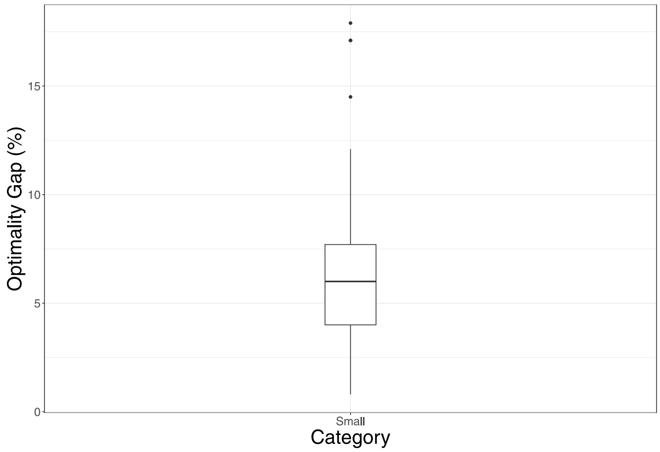


Fig. 6. The optimality gap for the small-scale instances with respect to the optimal solution.

Table 2
Effect of valid inequalities on the performance of the solver.

Inequality	Average improvement (%)	Standard error (%)	95% confidence interval (%)
Cost	40.8	3.1	34.8 - 46.9
Symmetry	8.4	0.8	6.8 - 9.9
Knapsack	3.6	0.3	3.0 - 4.2

We generate 96 instances with customer counts ranging from 6 to 150 and grid sizes ranging from 5 × 5 miles to 40 × 40 miles.

We extracted two performance measures from the experiments. The first is the cost savings obtained via the truck–drone approach as compared to the truck-only solution obtained from `solvePDP()` in Algorithm 1 (which we denote by *SPDP*). The second measure (*SL*) seeks to provide insight into labor-cost savings that are not explicitly captured in our objective function, which is a function of travel costs. For this, we compare the total number of trucks utilized in the truck–drone approach to those used in the truck-only scenario.

We introduce two performance measures, $SPDP$ and SL, used to evaluate the efficiency of the drone-assisted parcel delivery (DAPDP) heuristic compared to the truck-only case. They are calculated as follows:

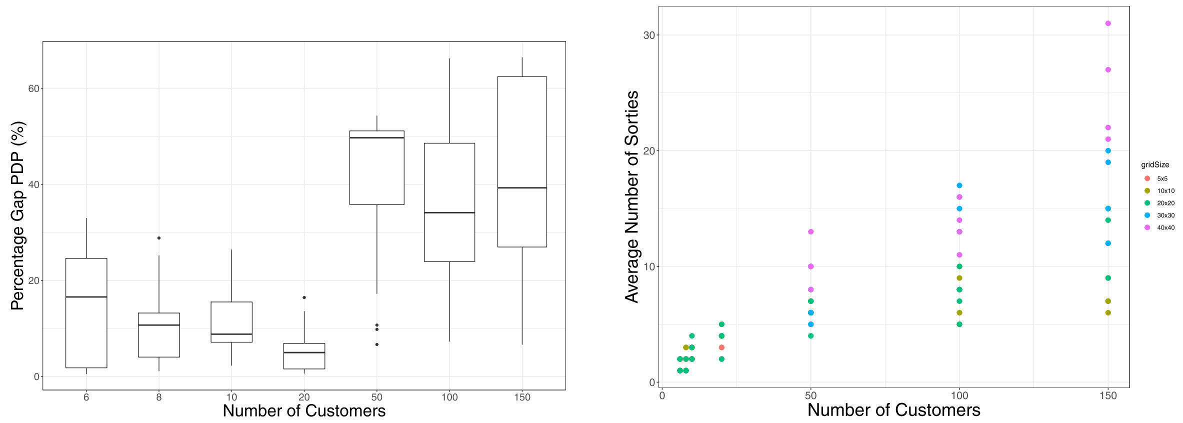
- $SPDP$ (representing some sort of "savings" in the PDP scenario) is equal to $1 - \frac{z^{DAPDP}}{z^{PDP}}$, where z^{DAPDP} is the best objective found using the truck–drone approach and z^{PDP} is the best objective found for the truck-only case. If the best objective found using the DAPDP heuristic is less than the best objective found in the PDP scenario, $SPDP$ will be positive, indicating savings or improvement.
- SL (representing some sort of "savings" in terms of resources, like labor or vehicles) is equal to $1 - \frac{\#V^{DAPDP}}{\#V^{PDP}}$, where $\#V^{DAPDP}$ is the number of trucks used in the truck–drone approach, and $\#V^{PDP}$ is the number of trucks used in the truck-only case (also referred to as PDP). If fewer trucks are used in the DAPDP scenario compared to the PDP scenario, SL will be positive, indicating savings or efficiency in resource utilization.

The values of $SPDP$ and SL are used as measures to evaluate the effectiveness of the DAPDP heuristic in terms of cost (the objective) and resource utilization (the number of trucks).

A comprehensive set of results for all 96 scenarios is presented in Table A.6 in Appendix A. From these results, we observe that the implementation of drones in parcel delivery can have a substantial impact. Specifically, drone-assisted parcel delivery can significantly improve the objective, which may include aspects such as delivery speed, efficiency, or cost, compared to a traditional truck-only delivery system. We also note that there is a reduction in the number of trucks deployed in at least 50% of the heuristic solutions in comparison to the truck-only case, as captured by the *SL* performance metric.

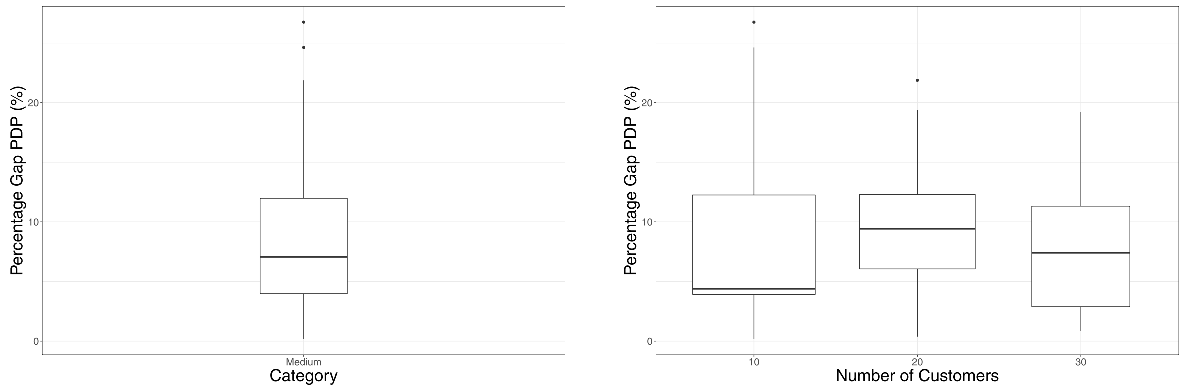
Fig. 7(a) quantifies the savings, which are 25% on average, in comparison to the truck-only case (PDP). However, we note that the degree of savings presents a larger variance in larger-scale instances. We explain this phenomenon by the fact that there are more opportunities to deploy the cheaper drone in instances with many customers in comparison to those with few customers. As observed in Fig. 7(b), which illustrates the average number of drone sorties in the solutions for each experimental scenario, the quantity of customers catered to by drones exhibits linear growth with an increase in the total number of customers. This corresponds to roughly half, or 50%, of the eligible customers that a drone can potentially serve.

This opportunity to deploy the drone is especially beneficial on a grid size of 10 × 10 miles and deteriorates for other grid sizes. This can be attributed to the particular drone parameters (i.e., speed and endurance), and we explore this sensitivity in Section 5.5.



(a) (b)

Fig. 7. Performance of the heuristic on large-scale instances. **(a)** The savings compared to the truck-only case; **(b)** The average number of sorties.



(a) (b)

Fig. 8. Summary of performance of the heuristic on benchmark instances. **(a)** DAPDP average saving (with respect to *m*-PDTSP); **(b)** Disaggregated according to the number of customers.

5.3. Benchmark instances

The closest benchmark to the DAPDP is the "Multi-Commodity One-to-One Pickup-and-Delivery Traveling Salesman Problem" (*m*-PDTSP of Hernández-Pérez and Salazar-González (2009)). Therefore, in this section, we investigate the savings in operational costs, if any, that the DAPDP offers.

Specifically, we are interested in the 210 "Class-3" instances where n random locations are generated on the square [−500, 500] × [−500, 500] that correspond to the customers with the single depot located at $(0, 0)$. The number of commodities is denoted by $m \in \{5,10,15\}$ which translates to {10,20,30} customers. The different capacities Q for each m, ∈ {500, 50, 45, … , 5} where each vertex represents the origin or destination of a unique commodity.

The boxplot in Fig. 8(a) summarizes the results, demonstrating an improvement with a median of about 8% and rising as high as 20%. The disaggregated results are shown in Fig. 8(b) and are consistent with those in Fig. 7(a).

5.4. Comparison against existing studies

To offer a holistic understanding of our heuristic's performance, we contrast it against recognized truck-plus-drone approaches from the literature, in particular, the approach of Murray and Chu (2015). The necessary modifications for this comparison include:

- Minimizing makespan rather than cost.
- Assuming the vehicle has unlimited capacity, thus adopting the TSP framework.
- Ignoring the precedence constraints.
- Restricting the MILP run-time to 30 min in a similar manner to Murray and Chu (2015).

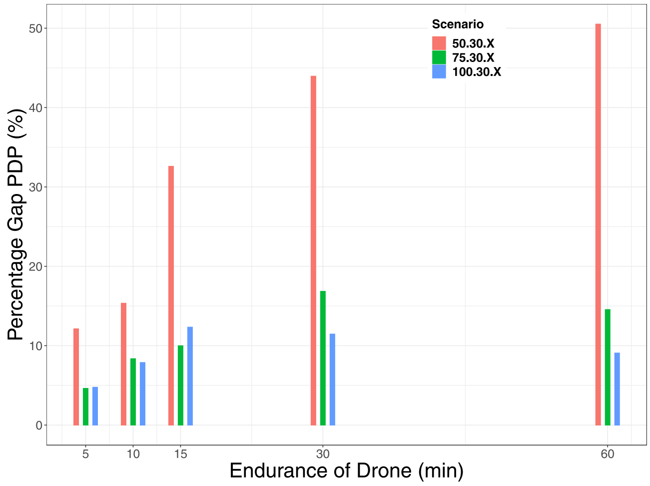


Fig. 9. The average saving in comparison to the truck-only case for different values of drone endurance.

Following Murray and Chu (2015), a set of 72 test problems was generated, each containing 10 customers randomly distributed across an 8-mile-square region. The depot location for each problem was randomly chosen to be either the average of the x- and y-coordinates of the customers, the average of the customers' x-coordinates with a y-coordinate of zero, or at the southwest corner of the region. Additionally, between 80% and 90% of the customers were designated as being UAV-eligible. The UAV's endurance was set at either 20 or 40 min. UAV speeds were selected as 15, 25, or 35 mph, with the additional assumption of Euclidean UAV flight paths. The corresponding truck speed was set to 25 mph. The service time parameters s_L and s_R as described in Table 1 were assumed to be one minute each. The optimality gap, which is the percentage difference between the heuristic and the solution obtained by CPLEX for the MILP formulation of the DAPDP, averages 0.4 percent with a standard error of 0.1 percent. This is within the confidence bounds of the performance reported in Murray and Chu (2015) for the TSP heuristic based on the savings criteria.

5.5. Sensitivity analysis

This section aims to investigate the relative significance of enhancing UAV flight speed, extending flight endurance, and reducing battery costs in order to determine their relative importance. Although the relative importance is determined primarily by the effect of these parameters on the objective, we also consider secondary effects such as the average count of UAV sorties and the percentage of customers eligible for UAV service that are actually served by UAVs. We vary each parameter of interest in Table 1 and use the experimental results to draw conclusions about the truck-UAV pickup and delivery problem.

We set the grid size to 30 × 30 miles to provide a clear comparison. The experimental setup comprises a suite of 120 arbitrarily constructed scenarios, divided into three different types, each including 40 distinct instances. The scenario types include "`50.30.X`", "`75.30.X`", and "`100.30.X`", following the format described in Section 5.2.

5.5.1. Endurance

Our base case in Table 1 sets the UAV's endurance time to 30 min. However, with ever-improving battery technology, we are interested in investigating the effect of a different endurance time on the objective. Additionally, we are interested in how much of the endurance time is consumed and how many UAV-eligible customers are actually visited as the endurance increases. It is not inconceivable that with a smaller fraction of endurance time consumed, perhaps the battery size and thus cost can also be reduced. Fig. 9 illustrates the influence of drone endurance on cost savings across various scenarios. As shown in the figure, the advantage derived from the synergy between trucks and drones widens with an increase in drone endurance. The operational cost can be reduced considerably with drones whose battery life lasts longer. A similar pattern can be discerned in Table 3, indicating a growth in the number of sorties correlating with increased drone endurance. Additionally, we see in Table 3 that the average fraction of endurance that is consumed per sortie decreases as the endurance of the drone increases.

Fig. 9 also shows that for a fixed drone endurance, the savings reduce with an increasing number of customers. This behavior is also evident in the experiments that follow, and we explain this shape as follows:

- Growth in the number of customers leads to an increased likelihood of pickup-delivery pairs that are geographically separated, given that the truck has to move for each sortie and package pickups are confined to trucks. Further, some deliveries have to be made by truck (owing to the capacity limitations of the UAV), which further reduces savings opportunities.
- When we increase the number of customers while holding other parameters constant, we are broadening the scope for drone usage. This is because there will be more potential customers within the operational range of the drones, and it paves the way for more optimal route planning where trucks cover less distance while drones make most of the deliveries.

Table 3
The average number of sorties as well as the average usage of total endurance time by those sorties across different scenarios and varying endurance times.

	Endurance	Scenario		
		50.30.X	75.30.X	100.30.X
Avg. Sorties	5 min	2.000	1.667	4.000
	7.5 min	4.750	4.000	9.000
	10 min	5.333	10.500	12.500
	15 min	5.500	9.333	14.667
	30 min	7.333	12.000	13.000
	60 min	7.000	7.500	16.000
(%) Endurance	5 min	77.904	80.944	81.921
	7.5 min	83.339	78.933	81.506
	10 min	77.632	81.433	79.970
	15 min	74.726	77.571	78.486
	30 min	73.887	68.445	70.385
	60 min	45.769	58.325	46.141

Table 4
The average number of sorties as well as the average usage of total endurance time by those sorties across different scenarios and varying UAV speeds.

	Speed	**Scenario**		
		50.30.X	75.30.X	100.30.X
Avg Sorties	25 mph	3.750	6.429	8.000
	35 mph	4.375	7.429	11.500
	50 mph	6.857	11.167	14.333
	75 mph	7.333	14.000	13.333
(%) Endurance	25 mph	85.180	84.546	81.359
	35 mph	79.151	82.432	84.739
	50 mph	73.598	65.116	66.677
	75 mph	63.101	61.559	69.299

5.5.2. Speed

The base case limits the UAV's speed to 50 miles per hour. However, we are curious about the effect of slower and faster speeds on the model's output because faster speeds allow us to reach more customers before the battery charge runs out.

Increasing the drone's speed, much like extending its endurance time, broadens the gamut of viable sorties. This is due to the fact that the drone, with its enhanced speed, can cover more distance and serve more customers within its battery life. As illustrated in Fig. 10, the benefit of the truck–drone collaboration grows as the UAV's speed increases. Furthermore, this benefit is greater than that observed in the endurance experiments depicted in Fig. 9. As a result, we conclude that faster-flying UAVs can significantly reduce operational costs. The same pattern is seen in the count of sorties in Table 4, which increases as the speed of the drone increases. Additionally, we can see in Table 4 that the average fraction of endurance that is consumed per sortie decreases as the endurance of the drone increases. Fig. 9 also shows that for a fixed drone speed, the savings reduce with an increasing number of customers. The reasons for this are the same as those for drone endurance detailed in Section 5.5.1.

To assess the robustness of our findings, we conducted a comprehensive sensitivity analysis of key drone parameters, including battery life, maximum payload, and drone operational costs. This analysis allows us to evaluate the performance of our proposed methodology and the overall benefits of UAV integration under different operational conditions and potential technological advancements. Our sensitivity analysis results indicate that our findings remain robust across a wide range of drone parameter values. Specifically, we observed that the cost savings and performance improvements achieved through the integration of UAVs into the pickup and delivery processes are generally maintained even when considering variations in drone parameters. This suggests that the benefits of incorporating UAVs into logistics operations are not limited to a narrow set of drone specifications but can be realized across a broad spectrum of drone capabilities. Additionally, the sensitivity analysis provides valuable insights into the relative importance of different drone parameters on the performance of UAV-assisted delivery systems. For example, we found that battery life and maximum payload have a greater impact on overall system performance than drone operational costs. This information can be useful for logistics operations managers and decision-makers when evaluating the trade-offs between different drone specifications and selecting the most appropriate drones for their specific operational requirements. In summary, our sensitivity analysis demonstrates the robustness of our findings and the potential for the integration of UAVs into pickup and delivery operations to yield significant cost savings and performance improvements across a wide range of drone parameter values.

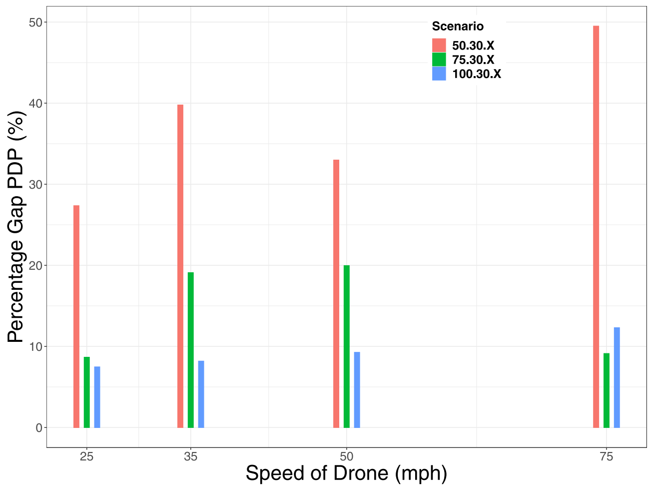


Fig. 10. The average saving in comparison to the truck-only case for different values of drone speed.

5.6. Managerial implications

Our study provides several key insights and actionable recommendations for logistics operations managers to enhance their understanding of the potential benefits and challenges of incorporating UAVs into their pickup and delivery operations. These implications can be summarized as follows:

1. Our findings demonstrate that integrating UAVs into the pickup and delivery process can lead to significant cost savings compared to traditional truck-only operations. This highlights the potential for UAVs to improve operational efficiency and reduce overall logistics costs.
2. The sensitivity analysis of drone parameters provides managers with an understanding of the factors that most significantly impact the performance of UAV-assisted delivery systems. These insights can help managers make informed decisions regarding drone selection, fleet size, and operational parameters.
3. Our proposed dynamic route and reassign heuristic can serve as a practical decision support tool for logistics operations managers, enabling them to quickly generate near-optimal solutions for large-scale DAPDP instances. This heuristic can be used to support strategic and tactical decision-making in the context of UAV-assisted delivery operations.
4. Our study also reveals the complementary nature of trucks and UAVs in pickup and delivery operations. Managers can use this information to more effectively allocate resources and create delivery strategies that take advantage of each vehicle type's advantages to increase efficiency and customer satisfaction.
5. Several use cases could benefit significantly from our proposed system:
    - Rural Deliveries: In areas where houses and establishments are spaced far apart, traditional truck deliveries often result in longer travel times and increased fuel consumption. Drones can quickly reach remote locations, ensuring timely deliveries without the need for the truck to cover the entire distance.
    - High Traffic Urban Areas: In dense urban areas, traffic congestion can delay deliveries. By using drones in tandem with trucks, packages can be delivered to their destination while the truck navigates the traffic, reducing overall delivery time.
    - Medical Supply Delivery: Especially in emergency situations, delivering medical supplies like blood or vaccines quickly can be vital. Drones can expedite these deliveries, reaching destinations faster than ground transportation, especially in congested or hard-to-reach areas like mountainous regions.
6. Finally, our research provides a foundation for further exploration and development of policies and regulations regarding UAV-assisted delivery systems. This includes addressing potential safety concerns and public acceptance, as well as the integration of these systems into existing transportation and logistics networks.

By incorporating these insights and recommendations into their strategic and operational decision-making processes, logistics operations managers can more effectively leverage UAV technology to optimize their pickup and delivery operations and achieve significant cost savings and performance improvements.

6. Conclusion

Last-mile delivery plays a prominent role in overall delivery times and customer satisfaction. Traditionally, in pickup and delivery vehicle routing problems, a truck departs a depot, picks up a package, and then delivers it to a customer on its route before making

Table A.5
The heuristic's performance on small-scale scenarios. $|C'|$ denotes the cardinality of the set of potential drone customers. The t^{MIP} column details the time taken (in seconds) to attain the optimal solution z^* using CPLEX, whereas t^{DAPDP} reports the time required by the heuristic to find its solution z^{DAPDP}. Opt. gap denotes the optimality gap.

Scenario	$\|C'\|$	z^*	t^{MIP} (s)	z^{DAPDP}	t^{DAPDP} (s)	Opt. gap
06.05.1	2	1.525	1	1.587	0.004	0.040
06.05.2	2	1.196	1	1.231	0.004	0.029
06.05.3	2	1.132	1	1.198	0.004	0.055
06.05.4	2	1.257	1	1.348	0.004	0.067
06.10.1	2	3.326	1	3.496	0.004	0.051
06.10.2	2	4.080	1	4.350	0.004	0.062
06.10.3	2	3.089	1	3.283	0.004	0.059
06.10.4	2	2.249	1	2.284	0.004	0.015
06.20.1	2	4.260	1	4.479	0.004	0.051
06.20.2	2	3.557	1	3.796	0.002	0.067
06.20.3	2	5.843	1	6.191	0.004	0.059
06.20.4	2	5.763	1	5.963	0.004	0.035
08.05.1	3	1.479	14	1.562	0.004	0.056
08.05.2	3	1.952	27	2.035	0.004	0.042
08.05.3	3	1.349	46	1.385	0.004	0.026
08.05.4	3	1.409	19	1.514	0.004	0.074
08.10.1	3	4.481	35	4.784	0.005	0.067
08.10.2	3	3.166	71	3.414	0.004	0.078
08.10.3	3	2.703	35	2.876	0.004	0.064
08.10.4	3	3.183	17	3.313	0.004	0.040
08.20.1	3	4.656	33	4.859	0.004	0.043
08.20.2	3	5.797	33	5.844	0.004	0.008
08.20.3	3	7.624	10	7.825	0.004	0.026
08.20.4	3	8.622	23	9.083	0.006	0.053
10.05.1	4	1.404	318	1.514	0.004	0.078
10.05.2	4	1.490	507	1.558	0.004	0.045
10.05.3	4	1.744	1002	1.852	0.006	0.061
10.05.4	4	1.528	395	1.646	0.004	0.077
10.10.1	4	3.620	1008	3.846	0.006	0.062
10.10.2	4	3.174	794	3.637	0.007	0.145
10.10.3	4	2.321	1002	2.717	2.100	0.171
10.10.4	4	2.715	543	2.761	0.006	0.016
10.20.1	4	4.684	921	4.992	0.008	0.065
10.20.2	4	5.933	1172	6.336	0.006	0.067
10.20.3	4	8.003	535	8.201	0.008	0.025
10.20.4	4	8.044	1002	8.238	0.006	0.024
12.05.1	5	1.921	60 231	2.146	0.050	0.117
12.05.2	5	1.530	14 932	1.668	0.040	0.090
12.05.3	5	1.999	4509	2.166	0.037	0.084
12.05.4	5	2.119	9071	2.200	0.049	0.038
12.10.1	5	4.277	11 626	4.797	0.081	0.121
12.10.2	5	3.535	16 365	3.839	0.046	0.086
12.10.3	5	3.788	34 140	4.081	0.063	0.077
12.10.4	5	3.808	45 361	4.462	0.063	0.171
12.20.1	5	4.866	21 736	5.123	0.064	0.053
12.20.2	5	8.908	46 952	10.504	3.052	0.179
12.20.3	5	8.663	44 514	8.761	0.051	0.011
12.20.4	5	5.912	4901	6.284	0.056	0.062

its way back to the depot. This problem is known as the "vehicle routing problem with pickup and delivery" (VRPPD). With the advent of drones and their ever-improving technology, the potential to reduce service times (and thus customer satisfaction) and minimize operational costs motivates their use in vehicle routing problems. Recent empirical studies underscore this notion. For instance, Merkert et al. (2022) explored consumer preferences for innovative and traditional last-mile parcel delivery options, highlighting that speed and timely deliveries substantially influence customer choices. Their findings suggest a growing inclination among consumers towards innovative delivery mechanisms, such as drones, primarily due to the perceived benefits of quicker and more predictable delivery times. Our study, by focusing on optimizing drone-assisted deliveries, resonates with these emerging consumer preferences and underscores the importance of advancing research in this domain.

This paper therefore introduces a novel variant of the truck-only problem. This variant incorporates a vehicle routing problem with pickup and delivery via drones, with the aim of minimizing costs. We denote this problem as the "drone-assisted pickup and delivery problem" (DAPDP) and propose an MILP formulation that is based on that presented in Murray and Chu (2015). We also propose an efficient heuristic for solving this problem. The performance analysis shows that the collaboration between trucks and drones can significantly improve the cost objective in comparison to the truck-only case. While our objective focuses on fuel costs, we include a mechanism for estimating labor cost reductions as well. Additionally, a sensitivity analysis via numerical experiments

Table A.6
Performance on each large-scale scenario. The table displays $|C'|$, representing the number of potential drone customers; $\#|C'|$, the average number of drone customers in the solution; $\#V^{PDP}$, the count of trucks used in the truck-only approach; $\#V^{DAPDP}$, the number of trucks involved in the truck–drone approach; *SL*, signifying the potential labor savings; t^{DAPDP}, indicating the time taken by the heuristic to find a solution; z^{PDP}, the best objective identified for the truck-only case; z^{DAPDP}, indicating the best objective found by the heuristic; and *SPDP*, the potential savings in operational costs.

Scenario	$\|C'\|$	$\#\|C'\|$	$\#V^{PDP}$	$\#V^{DAPDP}$	*SL* (%)	$t^{DAPDP}(s)$	z^{PDP}	z^{DAPDP}	*SPDP* (%)
6.5.1	2	1	1	1	0	1	2.469	2.112	14
6.5.2	2	1	1	1	0	1	2.360	2.349	1
6.5.3	2	2	1	1	0	1	2.222	2.191	1
6.5.4	2	1	1	1	0	1	1.407	1.145	18
6.10.1	2	2	1	1	0	1	3.201	2.304	28
6.10.2	2	1	1	1	0	1	3.324	2.228	32
6.10.3	2	2	1	1	0	1	3.066	3.049	1
6.10.4	2	1	1	1	0	1	3.401	2.476	27
6.20.1	2	2	1	1	0	1	8.513	8.325	2
6.20.2	2	1	1	1	0	1	5.155	5.059	1
6.20.3	2	1	1	1	0	1	5.835	4.672	19
6.20.4	2	2	1	1	0	1	6.233	4.757	23
8.5.1	3	2	1	1	0	1	2.463	2.242	8
8.5.2	3	2	1	1	0	1	1.511	1.451	3
8.5.3	3	2	1	1	0	1	2.310	1.644	28
8.5.4	3	1	1	1	0	1	2.722	2.383	12
8.10.1	3	3	1	1	0	1	3.685	3.645	1
8.10.2	3	1	1	1	0	1	1.965	1.763	10
8.10.3	3	3	1	1	0	1	3.956	3.344	15
8.10.4	3	2	1	1	0	1	4.396	3.905	11
8.20.1	3	1	1	1	0	1	5.632	4.214	25
8.20.2	3	1	1	1	0	1	9.113	7.985	12
8.20.3	3	1	1	1	0	1	8.519	8.177	4
8.20.4	3	2	1	1	0	1	7.631	7.506	1
10.5.1	4	2	1	1	0	1	2.942	2.877	2
10.5.2	4	2	1	1	0	1	2.723	2.281	16
10.5.3	4	2	1	1	0	1	2.377	1.749	26
10.5.4	4	2	1	1	0	1	2.214	2.052	7
10.10.1	4	3	1	1	0	1	4.964	4.616	7
10.10.2	4	2	1	1	0	1	5.026	4.533	9
10.10.3	4	3	1	1	0	1	3.803	3.200	15
10.10.4	4	3	1	1	0	1	4.933	4.469	9
10.20.1	4	2	1	1	0	1	6.360	5.955	6
10.20.2	4	2	1	1	0	1	10.123	9.400	7
10.20.3	4	3	1	1	0	1	8.452	7.152	15
10.20.4	4	4	1	1	0	1	7.812	7.172	8
20.5.1	9	2	1	1	0	1	3.552	3.531	1
20.5.2	9	4	1	1	0	1	3.306	2.763	16
20.5.3	9	3	1	1	0	1	3.194	2.994	6
20.5.4	9	3	1	1	0	1	3.030	3.007	1
20.10.1	9	5	1	1	0	1	5.558	5.231	5
20.10.2	9	4	1	1	0	1	3.753	3.243	13
20.10.3	9	4	1	1	0	1	6.155	5.818	5
20.10.4	9	4	1	1	0	1	6.315	6.209	1
20.20.1	9	5	1	1	0	1	11.658	11.531	1
20.20.2	9	4	1	1	0	1	12.060	11.602	3
20.20.3	9	4	1	1	0	1	14.442	13.800	4
20.20.4	9	2	1	1	0	1	12.003	10.957	8

demonstrates the relative importance of the drone parameters. These sensitivity results show that UAV parameters like endurance and speed have significant effects on the truck-UAV operation. The results show appreciable savings even when drones are operated at lower speeds.

Our findings support the conclusions of several previous studies, such as Murray and Chu (2015) and Ha et al. (2018), which demonstrated the potential benefits of integrating UAVs into logistics operations. Our results extend these findings by providing specific insights into the pickup and delivery problem in a drone-assisted context. In particular, our study shows that incorporating UAVs can lead to significant improvements in delivery efficiency and reductions in total travel time compared to truck-only operations. We also note that our heuristic finds utility when used as the procedure that determines an initial solution in metaheuristic methods, like that of Sacramento et al. (2019). Furthermore, our research findings align with recent industrial developments and practices. Major logistics companies such as Amazon, UPS, and DHL have been increasingly experimenting with drone-assisted delivery to enhance their operations (Ponza, 2016). Our study provides a rigorous framework for optimizing such drone-assisted pickup and delivery operations that can be applied in real-world settings. By comparing our findings with prior

Table A.7
Performance on each large-scale scenario (continued). The table displays $|C'|$, representing the number of potential drone customers; $\#|C'|$, the average number of drone customers in the solution; $\#V^{PDP}$, the count of trucks used in the truck-only approach; $\#V^{DAPDP}$, the number of trucks involved in the truck–drone approach; *SL*, signifying the potential labor savings; t^{DAPDP}, indicating the time taken by the heuristic to find a solution; z^{PDP}, the best objective identified for the truck-only case; z^{DAPDP}, indicating the best objective found by the heuristic; and *SPDP*, the potential savings in operational costs.

Scenario	$\|C'\|$	$\#\|C'\|$	$\#V^{PDP}$	$\#V^{DAPDP}$	*SL* (%)	$t^{DAPDP}(s)$	z^{PDP}	z^{DAPDP}	*SPDP* (%)
50.10.1	22	6	2	1	50	20	14.574	7.433	49
50.10.2	22	7	2	1	50	18	14.364	7.144	50
50.10.3	22	7	2	1	50	24	13.756	6.931	49
50.10.4	22	6	2	1	50	20	13.388	6.568	50
50.20.1	22	6	2	1	50	30	23.815	11.884	50
50.20.2	22	6	2	1	50	17	22.527	11.008	51
50.20.3	22	7	2	1	50	20	23.540	11.890	49
50.20.4	22	4	2	1	50	27	22.442	10.259	54
50.30.1	22	6	2	2	0	21	37.943	29.213	23
50.30.2	22	5	2	1	50	24	36.854	17.674	52
50.30.3	22	10	2	1	50	22	33.852	16.556	51
50.30.4	22	8	2	2	0	19	35.635	22.886	35
50.40.1	22	10	2	2	0	27	48.983	45.731	6
50.40.2	22	8	2	2	0	19	48.607	40.249	17
50.40.3	22	13	2	2	0	20	48.245	43.525	9
50.40.4	22	10	2	2	0	23	53.186	47.502	10
100.10.1	45	9	2	1	50	505	18.589	9.553	48
100.10.2	45	10	2	1	50	546	18.037	9.308	48
100.10.3	45	8	2	1	50	555	17.523	8.515	51
100.10.4	45	6	3	1	67	641	18.979	6.414	66
100.20.1	45	10	2	2	0	577	36.165	26.894	25
100.20.2	45	7	2	2	0	617	33.335	22.182	33
100.20.3	45	8	3	2	33	523	40.682	21.701	46
100.20.4	45	5	3	2	33	672	38.167	16.130	57
100.30.1	45	15	3	3	0	865	60.887	45.946	24
100.30.2	45	16	3	2	33	837	53.298	34.785	34
100.30.3	45	13	2	2	0	680	54.251	48.947	9
100.30.4	45	17	2	2	0	703	52.698	48.870	7
100.40.1	45	16	3	3	0	587	78.936	60.224	23
100.40.2	45	14	3	3	0	820	75.239	63.242	15
100.40.3	45	11	3	3	0	782	73.676	49.234	33
100.40.4	45	13	3	3	0	695	79.770	65.726	17
150.10.1	67	6	2	1	50	2491	24.337	8.219	66
150.10.2	67	7	2	1	50	1600	23.382	7.843	66
150.10.3	67	9	2	1	50	1988	23.328	8.768	62
150.10.4	67	7	2	1	50	1721	27.317	9.559	65
150.20.1	67	9	3	3	0	2559	44.019	24.821	43
150.20.2	67	12	3	2	33	2498	44.562	27.063	39
150.20.3	67	15	3	2	33	3324	49.468	27.746	43
150.20.4	67	14	3	3	0	2675	47.178	34.223	27
150.30.1	67	20	4	4	0	2658	73.717	50.357	31
150.30.2	67	12	4	4	0	3179	76.579	42.268	44
150.30.3	67	15	4	4	0	2635	73.942	47.598	35
150.30.4	67	19	4	4	0	3128	76.349	55.778	26
150.40.1	67	21	4	4	0	2452	95.342	83.994	11
150.40.2	67	27	4	3	25	2743	97.358	90.498	7
150.40.3	67	31	4	4	0	2652	93.205	75.182	19
150.40.4	67	22	4	4	0	2804	101.08	94.405	6

literature and investigating their implications for industrial practice, we believe we have strengthened the relevance and practical applicability of our study.

Many extensions of the DAPDP are possible. One such extension is an allowance for the formulation to consider multiple UAVs per truck. One additional complexity in this case is the determination of the optimal number of UAVs to deploy per truck. Another research track could consider the dynamic case where routes are determined in real-time (or as close to real-time as possible) in situations where there are cancellations or breakdowns of vehicles. Moving forward, future research endeavors could focus on devising exact strategies to solve the DAPDP optimally. Given that the DAPDP is a derivative of the VRP, its structure enables the application of Dantzig–Wolfe decomposition, which can provide a more precise formulation.

CRediT authorship contribution statement

Timothy Mulumba: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ali Diabat:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

I hereby declare that I have no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

I confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us. We understand that the Corresponding Author is the sole contact for the Editorial process. He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs. We confirm that we have provided a current, correct email address which is accessible by the Corresponding Author, and which has been configured to accept email from your journal.

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Appendix A. Results for random instances

Table A.5 presents the results from running the DAPDP heuristic on the 48 instances from the small-scale scenario in Section 5.2.1. The table reports t^{MIP}, which is the runtime (in seconds) in CPLEX to achieve the optimal solution z^*. The column t^{DAPDP} denotes the time taken by the heuristic to arrive at its solution, z^{DAPDP}.

Table A.6 and Table A.7 present the results from running the DAPDP heuristic on the 96 instances from the large-scale scenario discussed in Section 5.2.3. The table displays $|C'|$, representing the number of potential drone customers; z^{DAPDP}, indicating the best objective found using the truck–drone strategy; $\#|C'|$, the average number of drone customers in the solution; $\#V^{PDP}$, the count of trucks used in the truck-only approach; $\#V^{DAPDP}$, the number of trucks involved in the truck–drone approach; *SL*, signifying the potential labor savings; z^{PDP}, the best objective identified for the truck-only case; and *SPDP*, the potential savings in operational costs.

Appendix B. Google's OR-tools

Google's OR.Tools (2019) is a publicly available software suite designed to tackle vehicle routing problems. This suite features one of the most proficient VRP solvers, equipped with several heuristic approaches such as the Clarke–Wright savings heuristic (Clarke and Wright, 1964), Sweep heuristic (Wren and Holliday, 1972) and Christofides' heuristic (Christofides, 1976) for establishing an initial solution. Moreover, it employs metaheuristics like Guided Local Search (Voudouris and Tsang, 1999), Tabu Search (Glover and Laguna, 1998) and Simulated Annealing (Kirkpatrick et al., 1983) to avoid local optima.

One challenge with OR-Tools is the requirement of integer values for depot and customer locations, while our problem randomly allocates non-integer locations with three decimal places of precision. To circumvent this, we scale up the problem by inflating all locations by a factor of 10^3. After solving the problem, we shrink the solutions and tours to return to the original problem's dimensions.

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